The Halvorsen–Skogestad conjecture: the highest peak of the $V_{\min}$ diagram bounds the vaporization of every equilibrium-stage distillation arrangement

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Version 2 — Foundations, red team and integration: Claude Opus 5.5 (effort xhigh) · Sections: Claude Opus 5.5 (effort xhigh) · Network: Claude Opus 5.5 (effort xhigh) · Physical simulation: Claude Opus 5.5 (effort xhigh) · Independent verification: pending. Version 1 — Theory: Claude Opus 5.5 (effort xhigh) · Computation: Claude Opus 5.5 (effort xhigh) · General N: Claude Opus 5.5 (effort xhigh) · Independent verification: Claude Opus 5.5 (effort xhigh)

Abstract. Halvorsen and Skogestad (Ind. Eng. Chem. Res. 2003) conjectured that the generalized Petlyuk arrangement needs less vaporization than any other adiabatic distillation arrangement without internal heat exchange, for an ideal feed (constant relative volatilities and molar flows, constant pressure, infinitely many stages, pure products). We formalize "any arrangement" as the class $\mathcal{A}$ of all finite networks of constant-molar-overflow equilibrium stages, single-phase splitters and mixers, total condensers and total vaporizers, with one feed that may be split and conditioned, heat duties at any stage, recycles and products of either phase; infinitely many stages and pure products enter as a limit. We prove that at every steady state of every arrangement in $\mathcal{A}$, with any numbers of stages and any purities, the total heat supplied satisfies $R \geq \sum_P \max(0, \varphi_P(\theta_j))$ for each root θ_j of the feed equation, where $\varphi_P(\theta) = v_P - \sum_i \alpha_i n_{P,i}/(\alpha_i - \theta)$ is the Underwood flux of the product stream P; for products of purity $1-\varepsilon$ this gives $R \geq H_j - c_j\varepsilon$, and in the limit $R \geq \max_j H_j$, the highest peak of the $V_{\min}$ diagram, which is the minimum vapor of the generalized Petlyuk arrangement. The proof is elementary: every stream leaving an equilibrium stage has an Underwood flux of the sign of $-\sum_i \alpha_i x_i/(\alpha_i - \theta_j)$, x the stage liquid, so the units split into a light and a heavy side, and the energy balance of the heavy side gives the bound, with an exact expression of the excess as three nonnegative losses. It covers stream splitting, recycles, several feed stages, reverse flows, degenerate distributions, non-adiabatic sections and Murphree trays, which version 1 of this paper could not treat, and it shows that the five-component point below the Petlyuk value found in version 1 in the permissive form of Underwood's model cannot be approached by physical operation. The generalized Petlyuk arrangement attains the bound in Underwood's infinite-stage model (version 1) and, for two components, in the finite-stage model; for three or more components steady states exist at every prescribed flows, an isolated prefractionator is proved to admit exact finite-stage profiles, and the closed Petlyuk network reaches pure products at $R = \max_j H_j + \delta$ in three independent simulations, but its finite-stage attainment is not proved. Hence no arrangement of $\mathcal{A}$ needs less vaporization than the Petlyuk value computed by Halvorsen and Skogestad, and the conjecture is completely proved for two components. Version 1's results in the Underwood model, independently verified, are retained.

1 The question

I. J. Halvorsen and S. Skogestad, *Minimum energy consumption in multicomponent distillation. 3. More than three products and generalized Petlyuk arrangements*, Ind. Eng. Chem. Res. 42 (2003) 616–629 [1]. The abstract states:

> *"We also conjecture that the minimum energy requirement for the generalized extended Petlyuk arrangement is lower than the minimum-energy requirement for any distillation configuration when we consider conventional adiabatic sections and no internal heat exchange."*

Section 6.3 (p. 626) makes the assumptions explicit and records the status:

> *"We assume constant relative volatilities, constant molar flows, constant pressure, and no internal heat integration. Then the generalized adiabatic Petlyuk arrangement has the lowest need for vaporization compared to any other adiabatic distillation arrangement for separation of an arbitrary feed mixture into its pure components. We have not presented a complete proof, so the above conclusion is a conjecture."*

The number of stages is infinite; an adiabatic section is "a column section with constant molar flows and no heat exchange along the section" ([1], §6). Even the minimum vapor of the Petlyuk arrangement itself was left partly open (Section 5.3, p. 624):

> *"We have not carried out a detailed proof for the general N-component M-product case for other than the far left and right peaks."*

The conjecture is restated in [2], where infinitely many stages are assumed "because this provides a lower bound on the energy requirement for ideal mixtures", the energy is measured by the total vaporization, and the case with internal heat exchange is called unknown. For three components, Fidkowski and Królikowski [3] compared the Petlyuk column with side strippers, side rectifiers and the direct and indirect sequences in closed form; global MINLP studies [4] [5] find the Petlyuk value optimal on the instances they solve; we found no proof (novelty checks in `references.md`).

Version 1 of this paper (independently verified) proved the bound in Underwood's infinite-stage model for networks of two-section columns with interval streams; the verifier found that the model's permissive form admits a five-component point below the Petlyuk value, and listed what remained open: several feed stages, stream splitters, recycles, reverse net flows and degenerate operating points. Version 1 is kept as Sections 9–14.

Version 2 replaces the Underwood model by the physical model itself: constant-molar-overflow networks of equilibrium stages with any number of stages, of which "infinitely many stages and pure products" is a limit. In this model the lower bound has an elementary proof that needs no information on sections, roots or topology.

Theorem 1 (main results, version 2)**.** Let $\alpha_1 > \cdots > \alpha_N > 0$, $z_p > 0$, $\sum_p z_p = 1$, $q \in \mathbb{R}$, let $\theta_j \in (\alpha_{j+1}, \alpha_j)$ be the roots of $\sum_p \alpha_p z_p/(\alpha_p - \theta) = 1 - q$ and $H_j = \sum_{p>j} \alpha_p z_p/(\theta_j - \alpha_p)$.

(1) (Lower bound, every arrangement.) At every steady state of every arrangement of the class $\mathcal{A}$ (Definition 2: any finite network of equilibrium stages, single-phase splitters and mixers, total condensers and total vaporizers, one feed split and conditioned at will, heat duties at any stage, recycles, products of either phase), with any numbers of stages and any product purities, the total heat supplied R satisfies $R \geq \sum_P \max(0, \varphi_P(\theta_j))$ for every j, where φ_P is the

Underwood flux of the product stream P (Theorem 6); if every product has purity at least $1-\varepsilon$, then $R \geq \max_j(H_j - c_j\varepsilon)$ with an explicit c_j (Corollary 9). Hence every arrangement needs, in the limit of infinitely many stages and pure products, at least $\max_j H_j$.

(2) (Attainment.) The generalized Petlyuk arrangement attains $\max_j H_j$ in Underwood's infinite-stage model (Theorem 35, version 1). In the finite-stage model a single column attains H_1 for $N = 2$ (Proposition 10); for $N \geq 3$ the finite-stage Petlyuk minimum approaches $\max_j H_j$ in our simulations (Section 5). Finite-stage attainment of the closed Petlyuk network for $N \geq 3$ is not proved here.

Consequently no arrangement of $\mathcal{A}$ needs less total vaporization than the minimum vapor $\max_j H_j$ of the generalized Petlyuk arrangement computed by Halvorsen and Skogestad, the highest peak of the $V_{\min}$ diagram: the conjecture holds in $\mathcal{A}$ in this sense, for every N. For $N = 2$ it holds with the Petlyuk value also realized in the finite-stage model; for $N \geq 3$ the realization of $\max_j H_j$ as a finite-stage limit by the Petlyuk arrangement (Halvorsen and Skogestad's formula read in that sense) is supported numerically but not proved here (Section 5).

The class $\mathcal{A}$ contains every arrangement Halvorsen and Skogestad have in mind and more: non-adiabatic sections with external utilities, Murphree trays, any recycles. The proof (Section 3) rests on one observation: the Underwood flux $\varphi(\theta) = v - \sum_i \alpha_i n_i/(\alpha_i - \theta)$ of every stream leaving an equilibrium stage has the sign of $-\sum_i \alpha_i x_i/(\alpha_i - \theta)$, x the stage liquid, whatever the phase. At $\theta = \theta_j$ the units therefore split into a light side and a heavy side, and the energy balance of the heavy side, in Underwood's weights, gives the bound. It was found independently by three agents of version 2 (Section 8).

2 The class $\mathcal{A}$ and the physical model

The sources fix the physical assumptions: constant relative volatilities, constant molar flows, constant pressure, equilibrium stages, one feed of arbitrary composition and thermal state, pure products, and the total vaporization (heat input) as objective [1] [2] [6]. Halvorsen's thesis (§6.1.1) describes an arrangement as sections of connected equilibrium stages, reboilers and condensers, directly coupled by two-way liquid and vapor streams or coupled via condensers or reboilers, divided at stages with feed or side-draw streams; under constant molar overflow the energy balance of any stage reduces to "vapor out minus vapor in equals heat input" (*ibid.*, §2.6.1). We take the molar heat of vaporization as the unit of heat.

Definition 2 (the class $\mathcal{A}$)**.** An *arrangement* is a finite directed multigraph whose vertices are *units* and whose arcs are *streams*, together with a feed source and product outlets. A stream e has a phase (saturated liquid or saturated vapor), component flows $n_{e,i} \geq 0$ and vapor flow v_e ($= \sum_i n_{e,i}$ for a vapor, 0 for a liquid). The units are of four types:

(E) *equilibrium stage*: any inflows; an external heat duty $Q_u \in \mathbb{R}$; its contents leave as a liquid of composition x_u and a vapor of composition y_u with $y_{u,i} = \alpha_i x_{u,i}/S(x_u)$, $S(x) = \sum_k \alpha_k x_k$, each divided into any number of streams;

(M) *single-phase mixer or splitter*: inflows of one phase, outflows of that phase with the mixed composition, $Q_u = 0$;

(C) *total condenser*: vapor inflows, liquid outflows of the mixed composition, $Q_u = -\sum_{\text{in}} v_e$;

(R) *total vaporizer*: liquid inflows, vapor outflows of the mixed composition, $Q_u = +\sum_{\text{in}} \sum_i n_{e,i}$.

The process feed ($F = 1$, composition z, liquid fraction q) is divided into portions of sizes $f_u \geq 0$, $\sum f_u = 1$, entering units; every stream leaving the arrangement is a *product stream*, assigned to one component. A *steady state* satisfies, at every unit, the component balances $\sum_{\text{out}} n_{e,i} - \sum_{\text{in}} n_{e,i} = f_u z_i$, the energy balance $\sum_{\text{out}} v_e - \sum_{\text{in}} v_e = Q_u + (1-q) f_u$, and the composition rule of its type. Its *total vaporization* is $R = \sum_u \max(Q_u, 0)$, the total heat supplied.

Nothing is assumed on the topology (recycles are cycles of the graph), on the signs of the net component flows in a section, on which components are present where, on the numbers of stages, or on the stability or uniqueness of the steady state. An *adiabatic section* is a chain of (E) units with $Q = 0$, liquid flowing down and vapor flowing up with constant flows; a column is a stack of sections joined at junction stages. Every device of the arrangements considered in [1] [6] [4] [7] [8] [9] is a sub-network of units of $\mathcal{A}$ (Agent N's table, `total/network/network.md` §1.2): conventional columns with total or partial condensers and reboilers; thermal couplings; side draws of either phase; side strippers and rectifiers; Petlyuk, generalized Petlyuk and Kaibel arrangements and dividing-wall columns (a liquid or vapor split is a stage outflow divided into several streams, a merge is several inflows into one stage); several feed stages per column and intermediate sections; stream splitters dividing a section product among several columns, remixing of any streams, vapor or liquid bypasses, recycles; feed splitting, preheating, precooling and pre-flashing; intermediate reboilers and condensers, heaters and coolers at any stage (non-adiabatic sections with external utilities, which the conjecture's wording does not require us to admit); and trays with a Murphree vapor efficiency $E \in [0,1]$ common to all components, which are exactly an (E) unit receiving the fraction E of the vapor from below, a vapor bypass of the fraction $1-E$ and an (M) mixer ($Lx_{\text{in}} + EVy_{\text{in}} = Lx + EVK(x)x$), hence also packed sections as limits.

Definition 3 (infinitely many stages and pure products)**.** A steady state has *purity* at least $1-\varepsilon$ if every product stream assigned to component p has mole fraction of p at least $1-\varepsilon$. For a family $\mathcal{S} \subseteq \mathcal{A}$ (for example all stage numbers of one configuration) let $\Sigma_\varepsilon(\mathcal{S})$ be the set of steady states of members of $\mathcal{S}$ with purity at least $1-\varepsilon$, and put

$$R^*(\mathcal{S};\varepsilon) = \inf_{s \in \Sigma_\varepsilon(\mathcal{S})} R(s), \qquad R^*(\mathcal{S}) = \lim_{\varepsilon \to 0^+} R^*(\mathcal{S};\varepsilon).$$

The limit exists ($R^*(\mathcal{S};\varepsilon)$ is nonincreasing in ε). Halvorsen and Skogestad's minimum vapor of a configuration X with infinitely many stages is $R^*(\mathcal{S}_X)$, $\mathcal{S}_X$ the arrangements with the topology of X: every real arrangement has finitely many stages and impure products, and Underwood's values are the limits. In this language the conjecture reads: $R^*(\mathcal{S}_{\text{Petlyuk}}) \leq R^*(\mathcal{S})$ for every $\mathcal{S} \subseteq \mathcal{A}$.

Excluded, and why it is outside the conjecture. Heat exchange between process streams that is not counted in R ("internal heat exchange"; [1], §6.3 and §8; it lowers the external heat below the Petlyuk value, [6], Table 6.1); several pressures (excluded by constant pressure; they act through heat exchange between columns or compression, *ibid.*, §6.6, and change α); non-ideal mixtures and non-CMO enthalpies; devices releasing streams of different compositions not in phase equilibrium (decanters, membranes, component-dependent efficiencies); heat pumps; several process feeds of different compositions or thermal states. Finite stages and impure products are not excluded: they are the objects of the model.

3 The stage-level bound

For $\theta > 0$, $\theta \notin \{\alpha_i\}$, the *Underwood flux* of a stream e and of the unit feed are

$$\varphi_e(\theta) = v_e - \sum_i \frac{\alpha_i n_{e,i}}{\alpha_i - \theta}\,, \qquad \varphi_F(\theta) = (1-q) - \sum_i \frac{\alpha_i z_i}{\alpha_i - \theta}\,,$$

so that $\varphi_F(\theta_j) = 0$ for every j. For a section with vapor V and upward net flows w, the difference φ(vapor up) $-$ φ(liquid down) at any cross-section is $V - \sum_i \alpha_i w_i/(\alpha_i - \theta)$, the flux Λ_s of version 1. Write $G_x(\theta) = \sum_i \alpha_i x_i/(\alpha_i - \theta)$.

Lemma 4 (node balance). At every unit u of a steady state and every θ, product streams counting as outflows,

$$\sum_{e \text{ out of } u} \varphi_e(\theta) - \sum_{e \text{ into } u} \varphi_e(\theta) = Q_u + f_u \varphi_F(\theta).$$

Proof. φ is linear in (v, n): $\varphi_e = v_e - \sum_i c_i n_{e,i}$ with $c_i = \alpha_i/(\alpha_i - \theta)$. Subtract $\sum_i c_i$ times the component balances from the energy balance of Definition 2. □

Lemma 5 (sign uniformity). Let $\theta > 0$, $\theta \notin \{\alpha_i\}$. A liquid stream of flow l and composition x has $\varphi = -l\,G_x(\theta)$; a vapor stream of flow v in equilibrium with a liquid of composition x has $\varphi = -(\theta v/S(x))\,G_x(\theta)$. Hence all streams leaving an (E) unit have fluxes of one sign, that of $-G_{x_u}(\theta)$, and so do all streams leaving an (M), (C) or (R) unit (one phase, one composition).

Proof. For the vapor, with $\sum_i y_i = 1$ and $y_i = \alpha_i x_i/S$: $\varphi = v\sum_i y_i(1 - \frac{\alpha_i}{\alpha_i - \theta}) = -\theta v \sum_i \frac{y_i}{\alpha_i - \theta} = -\frac{\theta v}{S}\sum_i \frac{\alpha_i x_i}{\alpha_i - \theta}$, and $\theta > 0$, $S > 0$. □

Fix j and $\theta = \theta_j$. Call a unit *heavy-ish* if all its outflows have $\varphi \geq 0$ and *light-ish* otherwise; by Lemma 5 the outflows of a light-ish unit all have $\varphi < 0$ (or zero flow). Let $\mathcal{U}_+$ and $\mathcal{U}_-$ be the two sets, $R(\mathcal{W}) = \sum_{u\in\mathcal{W}} \max(Q_u, 0)$ and $Q(\mathcal{W}) = \sum_{u\in\mathcal{W}} \max(-Q_u, 0)$.

Theorem 6 (stage-level cut). At every steady state of every arrangement of $\mathcal{A}$ and for every j,

$$R - \sum_P \max(0, \varphi_P(\theta_j)) = R(\mathcal{U}_-) + Q(\mathcal{U}_+) + \sum_{e:\,\mathcal{U}_+\to\mathcal{U}_-} \varphi_e(\theta_j) - \sum_{e:\,\mathcal{U}_-\to\mathcal{U}_+} \varphi_e(\theta_j), \quad (7)$$

the sums running over the streams between the two sets, and every term on the

right is nonnegative. In particular $R \geq \sum_P \max(0, \varphi_P(\theta_j))$.

Proof. Sum Lemma 4 over $u \in \mathcal{U}_+$ at θ_j, where the feed terms vanish. Streams with both ends in $\mathcal{U}_+$ cancel, so $R(\mathcal{U}_+) - Q(\mathcal{U}_+) = \sum_{P \text{ from } \mathcal{U}_+} \varphi_P + \sum_{\mathcal{U}_+ \to \mathcal{U}_-} \varphi_e - \sum_{\mathcal{U}_- \to \mathcal{U}_+} \varphi_e$. Products leaving $\mathcal{U}_+$ have $\varphi_P \geq 0$ and products leaving $\mathcal{U}_-$ have $\varphi_P \leq 0$, so the first sum is $\sum_P \max(0, \varphi_P)$; streams leaving $\mathcal{U}_+$ have $\varphi \geq 0$ and streams leaving $\mathcal{U}_-$ have $\varphi \leq 0$, so the crossing terms are nonnegative. Add $R(\mathcal{U}_-) + Q(\mathcal{U}_+)$ to both sides. □

Corollary 8 (pure products)**.** If every product stream is pure, $R \geq H_j$ for every j; that is, $R \geq \max_j H_j$.

Proof. A pure liquid product of component p and flow P_p has $\varphi = \alpha_p P_p/(\theta_j - \alpha_p)$, a pure vapor product $\varphi = \theta_j P_p/(\theta_j - \alpha_p)$; both are positive exactly for $p > j$, and the vapor value is the larger. The products of component p carry z_p in total. □

Corollary 9 (finite purity)**.** If every product stream has purity at least $1 - \varepsilon$, then for every j

$$R \geq H_j - c_j \varepsilon, \qquad c_j = \max\Big(\frac{\alpha_j}{\alpha_j - \theta_j}, \frac{\alpha_{j+1}}{\theta_j - \alpha_{j+1}}\Big).$$

Consequently $R^*(\mathcal{S}) \geq \max_j H_j$ for every family $\mathcal{S} \subseteq \mathcal{A}$.

Proof. For a vapor stream, $\varphi^{\text{vap}} = \varphi^{\text{liq}} + (\text{flow})$ for the same component flows, so it suffices to bound the liquid fluxes. Keep only the products assigned to components $p > j$: $\sum_P \max(0, \varphi_P) \geq \sum_{i>j} \frac{\alpha_i n_i^h}{\theta_j - \alpha_i} - \sum_{i \leq j} \frac{\alpha_i n_i^h}{\alpha_i - \theta_j}$, n_i^h the flow of i in these products. By the overall balance $n_i^h = z_i - m_i$ for $i > j$, m_i the flow of i in products assigned to components $\leq j$. The flows m_i $(i > j)$ and n_i^h $(i \leq j)$ are disjoint parts of the total impurity, which is at most ε times the total product flow 1; the coefficients $\alpha_i/(\theta_j - \alpha_i)$ $(i > j)$ and $\alpha_i/(\alpha_i - \theta_j)$ $(i \leq j)$ are at most c_j. □

The proof uses only: constant molar overflow at every unit (Lemma 4); phase equilibrium with constant relative volatilities, or a single outflow composition and phase (Lemma 5); $\theta_j > 0$; one feed composition and thermal state, so that every feed portion has zero flux at θ_j; and the counting of every positive duty in R. Agents F, S, N and P checked the identities and the bound on finite-stage steady states computed by four independent solvers, and Agent N checked the identities in exact rational arithmetic (Section 6).

4 Consequences

The Underwood pinch hyperplane. For $0 \leq q \leq 1$ the equilibrium liquid x^F of the feed satisfies $G_{x^F}(\theta_j) = 0$ for every j (insert $z = (1-q)y^F + qx^F$ in the feed equation: $G_{x^F}(\theta_j)\,[(1-q)\theta_j/S(x^F) + q] = 0$). So a unit is light-ish or heavy-ish according to the side of the hyperplane $\{x : G_x(\theta_j) = 0\}$, through the feed-flash liquid, on which its liquid lies (for a (C) unit its liquid outflow; for an (R) unit the liquid in equilibrium with its vapor outflow, since $\sum_i y_i/(\alpha_i - \theta) = G_x(\theta)/S(x)$ for y in equilibrium with x). Theorem 6 says: *the heat supplied to the heavy side is at least* H_j (for pure products), and the excess of R over the bound is the sum of three nonnegative losses, heat supplied on the light side, heat removed on the heavy side, and flux across the hyperplane, as in pinch analysis of heat-exchanger networks.

Sections. In an adiabatic section, by Lemma 5 applied to two consecutive stages, $\Lambda_s(\theta) = L\,G_{x_{k+1}}(\theta) - (\theta V/S(x_k))\,G_{x_k}(\theta)$ at every cross-section (Underwood's identity, eq. (2.49) of [6] at the roots of Λ_s, here for every θ). Hence along a section the side changes at most once, from heavy-ish below to light-ish above if $\Lambda_s(\theta_j) > 0$ and the reverse if $\Lambda_s(\theta_j) < 0$, and a section whose stages change side contributes $|\Lambda_s(\theta_j)|$ to the defect (7). At a pinch composition $x^* \propto (w_i/(\alpha_i - \phi))_i$ of a section (ϕ a root of Λ_s), $G_{x^*}(\theta) = -(S(x^*)/V)\Lambda_s(\theta)/(\theta - \phi)$. (`total/sections/sections.md`, Lemmas 6–9; `total/network/network.md`, Corollary 6.2.)

Ties. By (7), a family of steady states with purity $\to 1$ and $R \to H_j$ must have, in the limit, no heat supplied to light-ish units, no heat removed from heavy-ish units, zero flux on every stream between the two sides, and θ_j as an Underwood root of every section whose stages change side, with the change taking place in a zone where the liquid is on the hyperplane: Underwood's pinch at an active root, now derived for every arrangement. In particular, at an optimum with a strict highest peak j^*, no reboiler or heater may deliver a light product $p \le j^*$ and no condenser or cooler a heavy product $p > j^*$ (Agent N, `network.md`, Corollary 5.3; version 1's Corollary 23 in the whole class). With finitely many stages the inequality is strict whenever some section changes side.

What the bound covers. Theorem 6 contains no hypothesis on reverse net flows (Halvorsen's regimes in which a component flows against the direction of a directly coupled section [6], §4.3.3), degenerate distributions, several feed stages, intermediate sections, stream splitters, remixing, recycles, bypasses, non-interval streams, products keeping every component of a column feed, or products of either phase: every gap of version 1 listed in its verification is closed for the lower bound. It also bounds the total vaporization of non-adiabatic arrangements with external utilities, such as reversible columns (whose total heat supply indeed exceeds the adiabatic Petlyuk value in Halvorsen's examples [6], Table 6.1), and of Murphree trays.

Internal heat exchange and several pressures. The theorem bounds the heat delivered to the heavy-ish units, whatever its origin. With heat exchange between process streams counted in R the bound still holds; the external heat can only fall below $\max_j H_j$ by moving heat into heavy-ish units from light-ish ones, which is what Halvorsen's heat-integrated examples do (side-stream condensation heating the feed, heat transfer across the dividing wall, double-effect columns). This is why the conjecture must exclude internal heat exchange, and it answers the question left in [2] for the total vaporization.

The permissive counterexample of version 1 is not physical. By Corollary 9, any family of steady states of configuration 161 (or of any arrangement) with purity $\to 1$ has $\liminf R \ge \max_j H_j$, while the permissive point of Section 13.3 has $R = H_3 - 2.6 \cdot 10^{-4}$. Concretely, its Underwood fluxes at θ_3 contain a light-to-heavy sign path C $\to$ CDE $\to$ BCDE $\to$ BCD $\to$ D, impossible at a steady state because along every arc of the sign graph a light-ish node can only be followed by a light-ish node (`total/network/network.md`, §6.3–6.4). The violation is global (a chain of four sections), which is why no column-local argument could exclude it. For global optimization this suggests a repair of the permissive model that keeps every physically admissible excess vapor: require, for every j, junction signs (+ at light products, − at heavy products) compatible with the signs of the section and link fluxes (one binary variable per junction and peak). Every point of the repaired model satisfies $R \ge H_j$, and limits of physical steady states satisfy its constraints when each junction of the Underwood description is

one equilibrium stage (*ibid.*, §6.5).

Version 1's strict theorem. The sign-graph argument of Theorem 22, which needed the column Lemmas 19 and 20 in the strict Underwood model, holds at every physical steady state with the partition $U = \mathcal{U}_-$: the node sign is a potential that never decreases along an arc of the sign graph.

5 Attainment

By Corollary 9 applied to the Petlyuk arrangement itself, $R^*(\mathcal{S}_{\text{Petlyuk}}) \geq \max_j H_j$; this is the part of Halvorsen and Skogestad's formula for the Petlyuk minimum that they proved only for the extreme peaks, now for every peak, every N and every q in the physical model. Since every arrangement satisfies the same bound, the conjecture $R^*(\mathcal{S}_{\text{Petlyuk}}) \leq R^*(\mathcal{S})$ for all $\mathcal{S}$ *follows from* the equality $R^*(\mathcal{S}_{\text{Petlyuk}}) = \max_j H_j$ (*attainment*: finite-stage Petlyuk steady states with purities tending to 1 and R tending to $\max_j H_j$), which is Halvorsen and Skogestad's formula for the Petlyuk minimum read in the finite-stage-limit sense. (Attainment is sufficient, not logically necessary: without it the conjecture could still hold, but the formula would fail.) In Underwood's infinite-stage model the generalized Petlyuk arrangement has an operating point with $R = \max_j H_j$, every internal column at its preferred split (Theorem 35 of version 1, independently verified, for every N and q; this is Halvorsen and Skogestad's construction [1]). In the finite-stage model:

Proposition 10 (two components; Agent S)**.** Let $N = 2$, $\alpha_1 = \alpha > 1 = \alpha_2$, light feed fraction $z \in (0,1)$, any q. For every $s > H_1$ and $\varepsilon > 0$ there is n_0 such that a column with $n_T, n_B \geq n_0$ stages, a partial reboiler of duty s and a total condenser has a steady state with both products of purity at least $1-\varepsilon$. Hence $R^*(\mathcal{S}_{\text{column}}) = H_1$, and for $N = 2$ the conjecture holds in $\mathcal{A}$.

Proof. Write $e(x) = \alpha x/(1+(\alpha-1)x)$ (light vapor fraction in equilibrium with light liquid fraction x), $D = z$, $B = 1-z$, $V_B = s$, $V_T = s+1-q$, $L_T = V_T - D$, $L_B = s+B$; $L_T > 0$ because $H_1 + 1 - q = \alpha z/(\alpha - \theta_1) > z$. The map $\beta_0(x) = (s/L_B)e(x)$ (bottom section, pure bottom product) has the fixed points 0 and $\hat{x}_B = (\alpha s/L_B - 1)/(\alpha - 1) \in (0,1)$ and $\beta_0(x) > x$ on $(0, \hat{x}_B)$; the map $\tau_0(x) = e^{-1}((L_T x + D)/V_T)$ (top section going down, pure top product) has the fixed points 1 and $\hat{x}_T = (V_T/L_T - 1)/(\alpha - 1)$ and $\tau_0(x) < x$ on $(\hat{x}_T, 1)$. Now $\hat{x}_T < \hat{x}_B$ iff $f(s) = \alpha s(s+1-q-z) - (s+1-q)(s+1-z) > 0$; after elimination of q by the feed equation, f is a quadratic with positive leading coefficient $\alpha - 1$ and roots H_1 and $s_- = 1/(\alpha-1) - z\theta_1/(\alpha - \theta_1) < H_1$ (`total/sections/scripts/check_binary_attainment.py`, sympy), so $\hat{x}_T < \hat{x}_B$ for $s > H_1$. For $\eta \in [0,1]$ put $x_B = \eta$ and $x_D(\eta) = 1 - (1-z)\eta/z$ (light balance). Iterating $x_{k+1} = (s\,e(x_k) + B\eta)/L_B$ upward from the reboiler liquid η ($n_B + 1$ steps) gives a feed-stage liquid $x_f^{\text{bot}}(\eta)$; iterating downward from the top vapor $x_D(\eta)$, $x = e^{-1}(y)$, $y_{\text{next}} = (L_T x + D x_D)/V_T$ ($n_T + 1$ stages) gives $x_f^{\text{top}}(\eta)$. Both are continuous, all iterates stay in $[0,1]$, and equality $x_f^{\text{bot}} = x_f^{\text{top}}$ yields a steady state (the feed-stage balance follows from the two cross-section balances and the overall balance). At $\eta = 0$, $x_f^{\text{bot}} = 0 < 1 = x_f^{\text{top}}$. Fix $\eta_1 \in (0,1]$ with $\eta_1 \max(1, (1-z)/z) \leq \varepsilon$. By monotonicity, $x_f^{\text{bot}}(\eta_1) \geq \beta_0^{n_B+1}(\eta_1) \to \hat{x}_B$ (or $\geq \hat{x}_B$) and $x_f^{\text{top}}(\eta_1) \leq \max(\hat{x}_T, \tau_0^{n_T}(e^{-1}(x_D(\eta_1)))) \to \hat{x}_T$, so for large n_T, n_B, $x_f^{\text{bot}}(\eta_1) > x_f^{\text{top}}(\eta_1)$, and the intermediate value theorem gives $\eta^* \in (0, \eta_1)$ with a steady state of purities $1-\eta^*$ and $1-(1-z)\eta^*/z$, both $\geq 1-\varepsilon$,

and $R = s$. □

For $N \geq 3$ the difficulty is not existence: steady states exist for every prescribed set of flows.

Proposition 11 (existence; Agent N)**.** Let an arrangement of $\mathcal{A}$ and the flows of all its streams and feed portions be prescribed, with the total mass balance at every unit, every unit reachable from a unit receiving feed and a product outlet reachable from every unit. Then for every $\alpha_1 > \cdots > \alpha_N > 0$ a steady state exists, and every steady state has all mole fractions in $[c, 1]$ for some $c > 0$ depending only on the flows, z and α_1/α_N. In particular R is determined by the flows, and $R^*(\mathcal{S}) \leq \rho$ as soon as, for every $\eta > 0$, some stage numbers and flows with $R \leq \rho + \eta$ admit a steady state with all purities at least $1 - \eta$.

Proof. (`total/network/network.md`, §10.) An (M) or (C) unit is an (E) unit without vapor outflow and an (M) vapor unit or an (R) unit one without liquid outflow, so the unknowns are one composition per unit. For fixed $K_{k,i} = \alpha_i / S(x_k)$ the balances of component i form a linear system whose matrix is a Z-matrix with positive diagonal, weakly chained diagonally dominant by columns (the excess of column k is the flow that unit k sends to product outlets, and every column is chained to such a unit), hence a nonsingular M-matrix with nonnegative inverse, positive between units joined by a path. Summing over i shows that $\sum_i x_{k,i} = 1$ holds automatically at every solution of the unnormalized system. Since $K_{k,i} \in [\alpha_N/\alpha_1, \alpha_1/\alpha_N]$, the positive entries of the inverses are bounded below, which gives the a priori bounds, uniformly along the homotopy $\alpha(t) = (\alpha_i^t)$; at $t = 0$ the system is linear with a unique nondegenerate solution, so the Brouwer degree is nonzero for every t. □

So attainment amounts to the *purity* of steady states at fixed flows. Isolated columns are understood. For $N = 2$ this is Proposition 10. For a directly coupled prefractionator with any number $m \geq 3$ of components and any feed quality, operated with its preferred net split deformed so as to keep its internal common roots and with a vapor margin η, there are exact, strictly positive finite-stage profiles whose unwanted extreme components at the two returns are at most $C\eta^{-1}\rho^n$, $0 < \rho < 1$; for $m = 3$ the major-component ratios of the two returns can moreover be prescribed (`total/attainment-astra/attainment.md`, Theorems 1–2, GPT-6-Astra; re-derived and re-implemented by Agent S, `total/sections/sections.md` §8.5). There the returns are boundary data: closing them through the finishing columns of the Petlyuk network, with every impurity controlled, is the missing step. The obstruction to a general proof is the coupling of pinch zones through the thermal couplings: in the infinite-stage limit, every section that carries excess vapor and is pinched at both ends has a composition front whose position is fixed only by exponentially small terms, so the pinch-matching limit system is not transversal there (Agent N's computation of the steady-state Jacobian: smallest singular value decreasing geometrically with n; Agent S's analysis of doubly pinched sections).

Numerical evidence. With the flows of Theorem 35 plus a circulation δ from the reboiler to the condenser spread over all sections, so that $R = \max_j H_j + \delta$ exactly, and n stages per section, the product impurities of the generalized Petlyuk arrangement fall as follows (Agent N, `total/network/scripts/ftc_fixed_flows.py`). For $N = 3$ ($\alpha = (4, 2, 1)$, $z_p = 1/3$, $q = 1$, $\delta = 0.002$) the largest impurity is $1.4 \cdot 10^{-2}$, $1.1 \cdot 10^{-3}$, $4.5 \cdot 10^{-7}$

and $1.3 \cdot 10^{-14}$ at $n = 20, 40, 80, 160$ (Newton residuals at most $2 \cdot 10^{-8}$); Agent P's independent solver, with a circulation $\delta = 0.01$ through both sections of every column, gives $8.1 \cdot 10^{-3}$, $1.1 \cdot 10^{-4}$ and $9.3 \cdot 10^{-9}$ at $n = 20, 40, 80$, a geometric decay of about 0.8 per stage, and Agent F's solver, with the prefractionator at its preferred split plus 0.02 on both flows and $R = \max_j H_j + 0.04$, gives $6.4 \cdot 10^{-3}$, $6.7 \cdot 10^{-5}$ and $4.5 \cdot 10^{-9}$ at $n = 20, 40, 80$ (`total/foundations/scripts/petlyuk_attain.py`). For $N = 4$ (one random instance, $\delta = 0.01$) N obtains $7.8 \cdot 10^{-2}$, $3.0 \cdot 10^{-3}$, $5.9 \cdot 10^{-5}$ and $2.1 \cdot 10^{-6}$ at $n = 10, 40, 80, 120$, with Newton residuals of only 10^{-5} to $5 \cdot 10^{-7}$, so these values are indicative. Minimizing R at fixed purity (Agent P, `total/simulation/`), the optimum approaches the finite-purity bound of Theorem 6 as n grows ($N = 3$, purity 0.99: $R - \text{bound} = 2.9 \cdot 10^{-2}$, $3.1 \cdot 10^{-4}$, $3.0 \cdot 10^{-8}$ at $n = 20, 40, 80$; Agent F's separate solver and optimizer reproduce $R = 1.372571$ at $n = 20$), and the bound tends to $\max_j H_j$ as the purity tends to 1. With the same fixed-flow construction, Agent F's solver also gives purities tending to 1 for $q = 0$, 0.5 and 1.3 (largest impurity $2.2 \cdot 10^{-5}$, $1.7 \cdot 10^{-5}$ and $1.4 \cdot 10^{-4}$ at $n = 40$, $R = \max_j H_j + 0.04$).

Status and route for $N \geq 3$. For $N = 3$, Agents S and N have reduced the closed network to a finite problem: at fixed flows with margins, the sharp limit fixes all net flows, the prefractionator feed-stage composition (Astra's vector) and the side-draw composition (pure B), but leaves the major-component ratios at the two coupling junctions free (the two neutral modes); at finite n these and the two trace net flows of the prefractionator (C upward, A downward) are fixed by exponentially small balances, giving four unknowns, to be handled by a Poincaré–Miranda or degree argument in exponentially rescaled variables with Astra's prefractionator theorem and binary shooting arguments for the two finishing columns (`total/network/network.md` §10, `total/sections/sections.md` §8.5–8.6). This argument is not complete; finite-stage attainment for $N \geq 3$ therefore remains open, although every computation supports it.

6 Computer verification of version 2

No step of the proofs of Sections 3–4 depends on a computation. Four independent codes (written separately by Agents F, S, N and P from the equations of Definition 2) check the identities and the bound on finite-stage steady states; paths are relative to the problem folder, and the environment is WSL Fedora with Python 3, numpy, scipy, sympy, mpmath, pyomo and ipopt.

- `total/network/scripts/unit_identities_exact.py` (exact rationals and sympy): Lemma 5 symbolically for $N = 2, \ldots, 6$, and the summed node balance (the cut identity for an arbitrary set of units) on random unit graphs with recycles, arbitrary subsets and rational θ. PASS.

- `total/network/scripts/section_identity_exact.py` (exact rationals): the identity $\Lambda_s(\theta) = L\, G_{x_{k+1}}(\theta) - (\theta V / S(x_k)) G_{x_k}(\theta)$ and the single change of side per section, on 400 random stage-to-stage profiles with net flows of both signs. PASS.

- `total/foundations/scripts/unitnet.py` (Agent F; floating point, Newton with analytic Jacobian): a binary column above and below its minimum vapor (below it the products are impure and the bound is attained to 10^{-11} with 80+80 stages, as the feed pinch predicts) and 396 random networks of (E), (M), (C) and (R) units with recycles, bypasses, split feeds, $q \in [-0.5, 1.5]$, $N = 2, \ldots, 5$ and products of

either phase and any purity: node balances to 10^{-12}, no sign-uniformity violation, $\min(R - \text{bound}) = -2.6 \cdot 10^{-14}$. ALL PASS (about 30 s).

- `total/sections/scripts/check_stage_cut.py` (Agent S): columns, three-product Petlyuk arrangements with 0–9 stages per section and random networks: identity residuals $\leq 4 \cdot 10^{-10}$, cut residuals $\leq 2 \cdot 10^{-11}$, no violation. PASS.

- `total/network/scripts/cut_bound_check.py` (Agent N): random section/junction networks and a ternary Petlyuk grid with 8–40 stages per section. PASS. `permissive_sign_path.py`: the light-to-heavy sign path of the permissive point (40 digits).

- `total/simulation/scripts/test_stage_bound.py` (Agent P, own solver with pseudo-transient continuation and Newton in logarithmic variables): random networks (2–5 components, sections of 0–11 stages, bypasses, recycles, split feeds, duties of either sign anywhere, liquid and vapor products, reverse net flows): 1131 steady states, no sign-uniformity violation, $\min(R - \text{bound}) = -2 \cdot 10^{-10}$ (the convergence tolerance of those steady states); `test_section_sign.py`: the single change of side per section on 3053 sections, no violation. PASS.

- Attainment runs (Section 5), each in the `scripts/` folder of its agent under `total/`:

  - N: `ftc_fixed_flows.py`;
  - P: `ftc_fixed_flows.py`, `attain.py`;
  - F: `petlyuk_attain.py`;
  - S: `check_binary_attainment.py`, `check_astra_prefrac.py`.

7 Scope, limitations and open questions

(1) *What is proved.* The lower bound $R \geq \sum_P \max(0, \varphi_P(\theta_j))$, hence $R^*(\mathcal{S}) \geq \max_j H_j$, for every arrangement of $\mathcal{A}$ (Theorem 6, Corollary 9); all statements of version 1 in the Underwood model (Sections 9–14), whose verification is in `verification.md`.

(2) *Attainment.* The equality $R^*(\mathcal{S}_{\text{Petlyuk}}) = \max_j H_j$, which implies the conjecture once the lower bound is known, is proved in Underwood's infinite-stage model for every N (Theorem 35) and in the finite-stage model for $N = 2$ (Proposition 10). For $N \geq 3$, steady states exist for all prescribed flows (Proposition 11), an isolated prefractionator admits exact finite-stage profiles with exponentially small traces, and the full network reaches purities tending to 1 at $R = \max_j H_j + \delta$ in three independent simulations; a proof for the closed network is not given (Section 5 describes the reduction for $N = 3$).

(3) *Class.* $\mathcal{A}$ contains every arrangement within the assumptions of [1]; the devices it excludes (heat exchange between process streams not counted in R, several pressures, non-ideal mixtures, decanters, membranes, component-dependent efficiencies, heat pumps, several feeds) are excluded by the conjecture itself. The bound concerns the total vaporization; with heat exchange between process streams the external heat can be lower (Section 4), and its minimum is not determined here.

(4) *Extensions.* For several feeds the same argument gives

$$R \geq \sup_{\theta} \Big[\sum_{P} \max(0, \varphi_P(\theta)) - \sum_{k} f_k \max(0, \varphi_{F_k}(\theta)) \Big],$$

and for non-sharp product specifications $R \geq \max_j \sum_P \max(0, \varphi_P(\theta_j))$ as it stands (`total/network/network.md`, §7); whether these bounds are attained is not studied. Constant molar overflow enters only through the energy balance: under the constant-heat-transport model of [10] (constant but component-dependent molar latent heats λ_i, no sensible heat), the same proof applies verbatim to the latent-heat-weighted flows $\tilde{n}_{e,i} = \lambda_i n_{e,i}$, because the weighted mole fractions satisfy $\tilde{y}_i = \alpha_i \tilde{x}_i / \sum_k \alpha_k \tilde{x}_k$ again, and bounds the total heat by the highest peak of the correspondingly weighted $V_{\min}$ diagram. Non-ideal mixtures are open: sign uniformity uses constant relative volatilities.

(5) *Version 1's permissive model.* The bound fails there for $N = 5$ (Section 13.3); version 2 shows that this point is not a limit of physical operations, so the 19 undecided permissive pairs no longer bear on the conjecture.

8 Credits for version 2

The stage-level bound (Lemmas 4, 5, Theorem 6) was found independently and within a few minutes by three agents of version 2: Agent N (network; derivation logged at 13:38), Agent S (sections; 13:45) and Agent F (foundations; 13:44–13:47); Agent P (physical simulation) re-derived the algebra. Agent F wrote the definition of $\mathcal{A}$ (with the limit definition of infinitely many stages and pure products), the red team (Murphree trays, feed phases, internal heat exchange) and the integration; Agent S the complete proofs with the explicit finite-purity constant and the section and junction lemmas (`total/sections/sections.md`); Agent N the decomposition of physical devices into units, the cut identity for arbitrary unit sets, the tie structure, the sign potential and the sign path of the permissive point, the existence theorem for prescribed flows and fixed-flow attainment numerics (`total/network/network.md`); Agent P the independent stage-by-stage simulator, the tests and the attainment study (`total/simulation/`); Agent S also proved the two-component attainment (Proposition 10) and checked the prefractionator theorem. All these agents: Claude Opus 5.5 (effort xhigh). The prefractionator attainment theorem quoted in Section 5 is due to GPT-6-Astra via Codex CLI (effort xhigh); the targeted novelty search of the stage-level argument (`total/literature/`) to Claude Opus 5.5 (effort xhigh). An early independent check of the lower bound was started in `total/verification-early/`. Independent verification of version 2: pending.

9 Version 1: setting of the Underwood model

Sections 9–16 reproduce version 1 of this paper, independently verified (`verification.md`), with short notes pointing to version 2. Version 1 summarized its results as follows.

Theorem 12 (main results of version 1)**.** Let R denote the total vapor generated by all heat exchangers of an arrangement.

(1) (Strict model, every N.) At every feasible operating point of every two-section column network (Definition 13), in particular of every regular-column configuration, $R \geq \max_j H_j$.

(2) (Permissive model.) The same inequality holds at every feasible point (a) for every regular-column configuration when $N \le 4$; (b) for every network and every N, for $j = 1$ and $j = N - 1$; (c) for the FTC configuration and every N, for all j; (d) for every regular-column configuration and every N, for every j such that no mixed stream (one containing j and $j + 1$) is produced by two columns, under a non-degeneracy assumption (N). It fails for one five-component configuration (Section 13.3).

(3) (Achievability.) The FTC configuration has an operating point, feasible in both models, with $R = \max_j H_j$.

Part 1 is Theorem 22, part 2 is Theorems 26, 30, 34 and 29, part 3 is Theorem 35. As statements about physics, parts 1 and 2 are superseded by Theorem 6; part 3 is the attainment used in Section 5.

9.1 Networks of two-section columns

A *stream* is an interval $[i, m] = \{i, \dots, m\}$ of components ($i \le m$); $[p, p]$ is the pure product p and $[1, N]$ the feed.

Definition 13 (two-section column network)**.** A network consists of the N pure-product nodes and a finite set of *columns*. A column c has a label $[i, m]$ ($i < m$) and a split $[i, k]/[l, m]$ with $i \le k < m$, $i < l \le m$ and $l \le k + 1$ (components $l, \dots, k$ distribute). It has one feed stage, between a top and a bottom adiabatic section of infinitely many stages. Its top section delivers the distillate $\mathrm{dist}(c) = [i, k]$ to one node with that label (a column labelled $[i, k]$, or the pure product i if $k = i$), and its bottom section delivers the residue $\mathrm{res}(c) = [l, m]$ to one node labelled $[l, m]$. Exactly one column, the *feed column*, is labelled $[1, N]$ and receives the process feed; every other column receives at least one stream. A node may receive any number of streams (from several producers, as a distillate or as a residue), and several columns may carry the same label. Each link is a thermal coupling (the vapor and liquid of the producing section are exchanged directly at the receiving node) or passes through a heat exchanger, and every node may carry an external heat duty: vapor generated (a reboiler or heater, counted in R) or vapor condensed (a condenser or cooler). Pure products are withdrawn as saturated liquid.

Labels shrink strictly along links, so a network is acyclic and every column is reached from the feed column. Not covered: columns with several feed stages (intermediate sections between feeds), stream splitters that divide one section product among several columns, recycles, non-interval intermediate streams, splits in which a product keeps every component of the column's feed (excluded by $k < m$ and $l > i$), streams delivered to a node whose label differs from the stream's label, and the reverse flows excluded in Section 9.2. (All of these belong to the class $\mathcal{A}$ of version 2 and are covered by Theorem 6.)

Regular-column configurations [7] [4] are the networks with at most one column per label, in which the distillate of $[i, m]$ is $[i, k]$ for the largest $k < m$ such that a column $[i, k]$ exists (or $k = i$), the residue is $[l, m]$ for the smallest such $l > i$, a stream produced by one column is transferred by a thermal coupling or by a condenser (distillate; TG:

a partial condenser removing the reflux, the stream leaving as saturated vapor) or a reboiler (residue; the stream leaving as saturated liquid), a stream produced by two columns is a side draw (thermal couplings), and a pure product is withdrawn through a total condenser or a reboiler, or as a liquid side draw between the two sections that produce it. There are 8, 152 and 6128 such configurations for $N = 3, 4, 5$ (transfer choices included). The *FTC* (generalized Petlyuk) configuration contains every mixture stream, all links being thermal couplings or side draws; it has one reboiler and one condenser.

9.2 The Underwood model

Constant relative volatilities, constant molar overflow, constant pressure, infinitely many stages in every section. For a column c with split $[i,k]/[l,m]$: net upward component flows $d_p \geq 0$ in the top section ($i \leq p \leq k$), net downward flows $b_p \geq 0$ in the bottom section ($l \leq p \leq m$), net feed $f_p = d_p[p \leq k] + b_p[p \geq l]$ (the feed z for the feed column, otherwise the sum of the incoming flows), vapor flows $V_T, V_B \geq 0$ with $V_T - V_B = U_c$, where U_c is the net vapor entering the feed stage: $1 - q$ for the feed column; otherwise the vapor brought by the incoming distillate links, minus the vapor drawn by thermally coupled residue links, plus the vapor generated minus the vapor condensed at the node. In a regular-column configuration with TG's conventions, a thermally coupled or side-draw distillate link brings V_T(producer), a condensed one $D = V_T - L_T$, a coupled residue link draws V_B(producer) and a reboiled one draws nothing. Put

$$g_T(x) = \sum_{p \leq k} \frac{\alpha_p d_p}{\alpha_p - x}\,, \qquad g_B(x) = -\sum_{p \geq l} \frac{\alpha_p b_p}{\alpha_p - x}\,.$$

For every $r \in \{l-1, \ldots, k\}$ there is a root $\theta_{c,r} \in [\alpha_{r+1}, \alpha_r]$ of the feed equation $\sum_p \alpha_p f_p/(\alpha_p - \theta_{c,r}) = U_c$, and a threshold vapor Y_T (with $Y_B = Y_T - U_c$) such that

$$\begin{gathered} g_T(\theta_{c,r}) \leq Y_T \quad (r = l-1,\ r = k), \qquad g_T(\theta_{c,r}) = Y_T \quad (l \leq r \leq k-1), \\ 0 \leq Y_T \leq V_T, \qquad 0 \leq Y_B \leq V_B. \end{gathered} \tag{14}$$

This is TG's model [4] (constraints (A18)–(A23) of the arXiv version, including the equalities at the interior roots which, TG note, are needed for the correct degrees of freedom); we call it the *permissive* model. (Nallasivam et al. [5] use a closely related formulation: an inequality $V_T \geq g_T(\theta)$ at every root together with explicit recovery-ordering constraints, instead of TG's equalities at the interior roots. The results below are proved for TG's formulation.) The *strict* model adds $Y_T = V_T$ whenever the column has an interior root ($l \leq k-1$): the column then runs exactly at the Underwood vapor of each of its interior roots, as an infinite-stage column does whenever the components $l, \ldots, k$ actually distribute (then these roots are active): "we may apply the common roots being in the range of volatilities for the components distributed to both ends. We denote these roots active roots. If we have any active roots then $V = V_{\min}$" (Halvorsen [6], §4.2.1; see also §3.2.3–3.2.4). The permissive model also admits columns carrying vapor above their threshold ("a column may be operated above its threshold vapor requirement to supply the vapor to another column" [4]), physically a vapor bypass or a finite-stage effect. Strict-feasible points are permissive-feasible.

Degenerate operating points (added by the verifier). Both models fix the split of each column through the network, so the strict model imposes $Y_T = V_T$ whenever two or more components are *nominally* distributing, whether or not they appear in both products.

At a point where every component $l, \ldots, k$ has a positive flow in both products the interior roots are active and the strict model is exactly the infinite-stage behaviour. At a degenerate point, where a nominally distributing component is absent from one product (e.g. $d_C = 0$ in a split ABCD→ABC/BCD), an infinite-stage column is governed by its actual distribution: Halvorsen counts a component as distributing only if it "appears in both products, or is exactly at the limit of becoming distributing" ([6], §3.2.4), so with fewer than two distributing components no root is active and the column may physically carry vapor above its Underwood value, which the strict model forbids; TG's equalities at the nominal interior roots are then not physically required either. Relabelling the split does not help in general, because the delivered stream would no longer carry the label of its receiver (Definition 13). Such points are not covered by Theorem 22; the theorems are statements about the model as defined. (The results that use only the sign rule, Lemma 19, rest on the inequality $V_T \geq g_T(\theta)$ at a common root, which holds at any infinite-stage operating point with nonnegative flows; we expect, but have not written out, their extension to degenerate points. Version 2 settles degenerate points physically: Theorem 6.)

At the roots, $g_T - g_B = U_c$, hence $g_B(\theta_{c,r}) \leq Y_B$ for $r = l-1, k$ with equality for interior r.

Standing hypotheses. (H1) No reverse flows: $d_p \geq 0$ and $b_p \geq 0$ in every section. This is part of the model of [4] [5]: the conditions (14) express Underwood's root interlacing for top sections with $w \geq 0$ and bottom sections with $w \leq 0$, and with reverse flows the roots can leave their intervals (Halvorsen [6], §4.3.3–4.3.4). Reverse flows do occur in directly coupled columns operated away from the preferred split, e.g. in a Petlyuk prefractionator whose top liquid carries a component that the top section would otherwise remove (*ibid.*, §4.3.3), so (H1) is a genuine restriction of the operating points, not only of the configurations. The cut identity below needs no sign assumption, but the column lemmas use (H1) through the monotonicity of the Underwood functions; whether reverse flows could lower R is not addressed here (version 2: they cannot, Section 4). (H2) A single feed, and pure products withdrawn as saturated liquid.

The objective is R, the vapor generated by all heat exchangers (reboilers on products and on transfer streams, heaters at nodes).

9.3 Peaks and regions

Fix $j \in \{1, \ldots, N-1\}$ and write $\theta = \theta_j$. A stream, column or node with label $[i, m]$ is *light* if $m \leq j$, *heavy* if $i \geq j+1$, *mixed* if $i \leq j < m$. Producers of a mixed column are mixed (their labels contain its label); the distillate of a mixed column is light or mixed (it contains $i \leq j$), its residue mixed or heavy; the sections of a light (heavy) column join light (heavy) nodes. A *light exit* is the top section of a mixed column whose distillate is light; a *heavy exit* is the bottom section of a mixed column whose residue is heavy.

10 The Underwood flux and the cut identity

For a section s with vapor flow V_s and upward net component flows w_s ($w = d$ in a top section, $w = -b$ in a bottom section), define the *Underwood flux*

$$\Lambda_s(x) = V_s - \sum_p \frac{\alpha_p w_{s,p}}{\alpha_p - x}\,.$$

For a column c: $\Lambda_{c,T} = V_T - g_T$, $\Lambda_{c,B} = V_B - g_B$, and its *feed function* $\Lambda_{c,F}(x) = U_c - \sum_p \alpha_p f_p/(\alpha_p - x)$ satisfies $\Lambda_{c,T} - \Lambda_{c,B} = \Lambda_{c,F}$.

Lemma 15 (monotonicity)**.** On any interval free of poles, $\Lambda_{c,T}$ and $\Lambda_{c,F}$ are nonincreasing and $\Lambda_{c,B}$ is nondecreasing; $\Lambda_{c,F}$ is strictly decreasing if some $f_p > 0$.

Proof. The derivatives are $-\sum \alpha_p d_p/(\alpha_p - x)^2$, $-\sum \alpha_p f_p/(\alpha_p - x)^2$ and $+\sum \alpha_p b_p/(\alpha_p - x)^2$. □

10.1 Nodes and the identity

The *nodes* of a network are its columns and its pure products: a column node gathers the feed stage of the column, the heat exchangers on its incoming links and its external duty; a product node gathers the withdrawal point and its heat exchanger (or the side draw). Every section joins the node of its column to the node that receives its distillate (top section) or its residue (bottom section). We orient sections upward and write $s\colon X \to Y$ when the bottom end of s is in X and its top end in Y.

Proposition 16 (cut identity)**.** Let U be a set of nodes containing every light pure product and no heavy pure product, U^c its complement. Then

$$R - H_j = R(U) + Q(U^c) + \sum_{s\colon U^c \to U} \Lambda_s(\theta_j) - \sum_{s\colon U \to U^c} \Lambda_s(\theta_j), \tag{17}$$

where $R(U)$ is the reboiler duty (vapor generated) located in U and $Q(U^c) \geq 0$ the condenser duty (vapor condensed) located in U^c.

Proof. $\Lambda_s(x)$ is linear in (V_s, w_s). At a node, constant molar overflow and the absence of heat exchange give conservation of the total vapor flow and of the net component flows, except for: the feed, which injects vapor $1 - q$ and components z; a heat exchanger, which converts liquid into vapor (reboiler, duty Q_r) or vapor into liquid (condenser, duty Q_c); a liquid product withdrawal of z'_p of component p. A section whose top end is at the node brings the flux Λ_s into it; a section whose bottom end is at the node takes Λ_s out of it (its vapor leaves upward, its liquid arrives, its net flow w_s leaves upward). Hence, at every node,

$$\sum_{s \text{ ending at the node}} \Lambda_s(x) - \sum_{s \text{ starting at the node}} \Lambda_s(x) = Q_c - Q_r - \Lambda^{\text{feed}}(x) - \frac{\alpha_p z'_p}{\alpha_p - x},$$

with $\Lambda^{\text{feed}}(x) = (1 - q) - \sum_p \alpha_p z_p/(\alpha_p - x)$ at the feed column only and the last term at product nodes only. (Agent C verified this identity in exact rational arithmetic on all configurations for $N \leq 5$, with arbitrary junction duties.) At $x = \theta_j$, $\Lambda^{\text{feed}} = 0$ by the definition of θ_j. Sum the balances over the nodes of U^c: sections inside U^c cancel, every heavy product withdraws $\alpha_p z_p/(\theta_j - \alpha_p)$, in total H_j, and no light product lies in U^c. This gives $R(U^c) = H_j + Q(U^c) + \sum_{U^c \to U} \Lambda_s - \sum_{U \to U^c} \Lambda_s$; add $R(U)$. □

The canonical choice $U =$ light streams gives

$$R - H_j = R(\text{light}) + Q(\text{mixed}) + Q(\text{heavy}) + \sum_{\text{light exits}} \Lambda_s(\theta_j), \tag{18}$$

and $U =$ light $\cup$ mixed streams gives $R - H_j = R(\text{light}) + R(\text{mixed}) + Q(\text{heavy}) + \sum_{\text{heavy exits}} \Lambda_s(\theta_j)$. So $R \geq H_j$ as soon as the right-hand side of (17) is nonnegative

for one admissible U; conversely, the inequality is tight on open sets of parameters for many configurations (Agent C's computations), so no strict inequality can be expected.

11 Column lemmas

For a mixed column c put $A_c = \Lambda_{c,T}(\theta_j)$, $B_c = \Lambda_{c,B}(\theta_j)$ and $a_c = \Lambda_{c,F}(\theta_j) = A_c - B_c$.

Lemma 19 (sign rule)**.** For every mixed column, in both models: $A_c \geq \min(0, a_c)$, equivalently $B_c \geq \min(0, -a_c)$. In particular A_c and B_c are never both negative, $a_c \geq 0 \Rightarrow A_c \geq 0$ and $a_c \leq 0 \Rightarrow B_c \geq 0$. Moreover, if $k < j$ then $A_c \geq 0$, and if $l > j + 1$ then $B_c \geq 0$.

Proof. Let the split be $[i, k]/[l, m]$; (14) and $Y \leq V$ give $\Lambda_{c,T}(\theta_{c,r}) \geq 0$ and $\Lambda_{c,B}(\theta_{c,r}) \geq 0$ for $r \in \{l - 1, \ldots, k\}$. The values are finite: if $\theta_{c,r}$ is an endpoint $\alpha_{r'}$, the feed equation forces $f_{r'} = 0$, hence $d_{r'} = b_{r'} = 0$.

Case $l - 1 \leq j \leq k$. If all flows vanish the claim is trivial. Otherwise $\Lambda_{c,F}$ is strictly decreasing on $[\alpha_{j+1}, \alpha_j]$ (where finite) and vanishes at $\theta_{c,j}$. If $a_c \geq 0$ then $\theta_j \leq \theta_{c,j}$, and $A_c \geq \Lambda_{c,T}(\theta_{c,j}) \geq 0$ by Lemma 15. If $a_c \leq 0$ then $\theta_j \geq \theta_{c,j}$ and $B_c \geq \Lambda_{c,B}(\theta_{c,j}) \geq 0$, so $A_c = a_c + B_c \geq a_c$.

Case $k < j$. The top section contains only components $p \leq k$, so $\Lambda_{c,T}$ has no pole in $(-\infty, \theta_{c,k}]$; since $\theta_j < \alpha_j \leq \alpha_{k+1} \leq \theta_{c,k}$, $A_c \geq \Lambda_{c,T}(\theta_{c,k}) \geq 0$; and if $a_c \leq 0$, $B_c = A_c - a_c \geq 0$. *Case* $l > j + 1$: symmetric, with $\theta_{c,l-1} < \alpha_{j+1} < \theta_j$ and $\Lambda_{c,B}$ nondecreasing on $[\theta_{c,l-1}, \infty)$. □

Lemma 20 (strict model: interior split)**.** In the strict model, if both the distillate and the residue of a mixed column are mixed ($l \leq j \leq k - 1$), then A_c and B_c are never both positive.

Proof. j is an interior root of c, so $V_T = Y_T = g_T(\theta_{c,j})$: $\Lambda_{c,T}(\theta_{c,j}) = \Lambda_{c,B}(\theta_{c,j}) = 0$. If $\theta_j \leq \theta_{c,j}$ then $B_c \leq 0$; if $\theta_j \geq \theta_{c,j}$ then $A_c \leq 0$ (Lemma 15). □

For the feed column $a_{[1,N]} = \Lambda^{\text{feed}}(\theta_j) = 0$, so $A = B \geq 0$ by Lemma 19. *Propagation.* By the node balance of Proposition 16, a column c other than the feed column has

$$a_c = \sum_{P:\ \mathrm{dist}(P)=c} A_P - \sum_{P':\ \mathrm{res}(P')=c} B_{P'} + \rho_c - \kappa_c, \tag{21}$$

where $\rho_c \geq 0$ is the vapor generated and $\kappa_c \geq 0$ the vapor condensed at the node of c (in TG's convention: $\kappa_c = L_T(P)$ for a condensed distillate link, $\rho_c = V_B(P')$ for a reboiled residue link, 0 for couplings and side draws).

12 The strict model: all *N*

Theorem 22. In the strict model, every feasible operating point of every two-section column network satisfies $R \geq H_j$ for every j.

The argument is due to the computation agent (Agent C) and was checked by the theory agent; a standalone machine check of its combinatorial core is described in Section 15.

Proof. Fix j and an operating point. Build the *sign graph*: for every section $s : x \to y$, an arc $x \to y$ if $\Lambda_s(\theta_j) > 0$ and an arc $y \to x$ if $\Lambda_s(\theta_j) < 0$. Let U be the set of nodes

reachable from the light nodes (light nodes included). If U contains no heavy node, then every section $s: U^c \to U$ has $\Lambda_s \geq 0$ (otherwise its bottom end would be reachable) and every section $s: U \to U^c$ has $\Lambda_s \leq 0$; so all terms of (17) are nonnegative and $R \geq H_j$.

We show by induction on the length of a shortest path from the light nodes that every reached mixed node y satisfies (i) $B_y \geq 0$ and (ii) $A_y \leq 0$ if $\mathrm{dist}(y)$ is mixed, and that no heavy node is reached. The sections incident to a mixed column node y are its top section (to the node receiving $\mathrm{dist}(y)$), its bottom section (from the node receiving $\mathrm{res}(y)$) and the sections of its producers ending at y; we write $\mathrm{dist}(y)$, $\mathrm{res}(y)$ also for these nodes. Let y be reached by an arc from an already reached node z.

- $z = \mathrm{dist}(y)$ (light or mixed), arc down the top section of y: then $A_y < 0$, so $B_y \geq 0$ by Lemma 19; (i) and (ii) hold.
- $z = \mathrm{res}(y)$, arc up the bottom section of y: then $B_y > 0$. The node z is mixed (a residue of a mixed column is not light, and no heavy node has been reached). If $\mathrm{dist}(y)$ is mixed, both products of y are mixed and Lemma 20 gives $A_y \leq 0$. So (i) and (ii) hold.
- z a producer of y (necessarily mixed): an arc up the top section of z needs $A_z > 0$, excluded by (ii) for z since $\mathrm{dist}(z) = y$ is mixed; an arc down the bottom section of z needs $B_z < 0$, excluded by (i) for z.

A heavy node y could only be entered from a mixed column c with $\mathrm{res}(c) = y$, down the bottom section of c, which needs $B_c < 0$, against (i); no section joins a light node to a heavy node. Hence U contains no heavy node. □

Corollary 23 (necessary condition for a tie; Agent C)**.** In the strict model, every feasible point of every network satisfies $R - H_j \geq R(\mathrm{light}) + Q(\mathrm{heavy})$ for every j, where $R(\mathrm{light})$ is the vapor generated at nodes with labels inside $[1, j]$ and $Q(\mathrm{heavy})$ the vapor condensed at nodes with labels inside $[j+1, N]$. Hence, if H_{j^*} is the strict maximum of the peaks, a network can attain V_{FTC} only if it has no reboiler or heater with positive duty at a node inside $[1, j^*]$ and no condenser or cooler with positive duty at a node inside $[j^*+1, N]$.

Proof. In the proof of Theorem 22, U contains the light nodes, U^c the heavy ones, and all flux terms of (17) are nonnegative; so $R - H_j \geq R(U) + Q(U^c) \geq R(\mathrm{light}) + Q(\mathrm{heavy})$. □

In Agent C's runs ($N = 4$, $q = 1$), 2717 of 2720 observed ties satisfy this condition; the three exceptions are near-ties (relative gap $\sim 10^{-9}$) caused by trace components.

Remark 24. The proof uses only Lemmas 19–20 column by column, the interval labels and the identity (17). Heat duties never create arcs and enter (17) with nonnegative signs, which is why the theorem holds for the whole class of Definition 13: any thermal state of the transferred streams, heaters or coolers at any node, side draws merging any number of links, several columns with the same label, any q, and no regularity rule. Hypothesis (H1) enters through Lemma 15. ⌟

13 The permissive model

In the permissive model Lemma 20 fails: a column with an interior root may run above its threshold, with $A_c = B_c = V_T - Y_T > 0$. The reachability argument then leaves possible "bad paths"; Agent C's exact enumeration finds none for $N = 3$, exactly one

family for $N = 4$ (family K below), 70 paths in 48 basic configurations for $N = 5$ and 5795 in 2300 for $N = 6$.

13.1 Extreme peaks and paths

Lemma 25 (path). In a network in which every mixed column has at most one mixed child (the node receiving its distillate or its residue), $R \geq H_j$ in both models.

Proof. Every mixed column is reached from the feed column through mixed producers, and each has at most one mixed child; hence the mixed columns form a path $c_0 \to c_1 \to \cdots \to c_n$ (c_0 the feed column), and c_{t+1} has the single producer c_t. Write a_t, A_t, B_t for c_t, $\rho_t, \kappa_t \geq 0$ for the vapor generated and condensed at the node of c_{t+1}, T_t for the sum of A over the light exits among $c_t, \ldots, c_n$ and $K_t = \sum_{t' \geq t} \kappa_{t'}$. We claim $T_t \geq \min(0, a_t) - K_t$. For $t = n$: $T_n = A_n \geq \min(0, a_n)$ (Lemma 19; the top section of c_n is a light exit). If $c_{t+1} = \mathrm{dist}(c_t)$: $T_t = T_{t+1}$ and $a_{t+1} = A_t + \rho_t - \kappa_t \geq \min(0, a_t) - \kappa_t$. If $c_{t+1} = \mathrm{res}(c_t)$, the top of c_t is a light exit and $a_{t+1} = -B_t + \rho_t - \kappa_t$; then $T_t = A_t + T_{t+1} \geq A_t + \min(0, a_{t+1}) - K_{t+1}$, which is $\geq A_t - K_{t+1} \geq \min(0, a_t) - K_t$ if $a_{t+1} \geq 0$, and $\geq A_t - B_t - \kappa_t - K_{t+1} = a_t - K_t$ otherwise. With $a_0 = 0$: $\sum_{\text{light exits}} \Lambda_s = T_0 \geq -K_0 \geq -Q(\text{mixed})$, and (18) gives $R \geq H_j$. □

Theorem 26 (extreme peaks). In both models, for every N and every two-section column network, $R \geq H_1$ and $R \geq H_{N-1}$.

Proof. For $j = 1$ the mixed columns are labelled $[1, m]$, $m \geq 2$; their residues are heavy, so each has at most one mixed child (its distillate) and Lemma 25 applies. For $j = N - 1$ the mixed columns are labelled $[i, N]$ and their distillates are light. □

For $N = 3$ both peaks are extreme: the conjecture holds in both models for all eight ternary configurations and every q. For any N, the conjecture holds in both models whenever the highest peak is H_1 or H_{N-1}.

13.2 Recovery ordering and merge-free mixed networks

The results of this subsection and their proofs are due to the general-N agent (Agent G, `general/general-N.md`); they hold for regular-column configurations under the non-degeneracy assumption

> (N) every column receives a positive net flow of every component of its label.

Lemma 27 (recovery ordering). Let c be a column with split $[i, k]/[l, m]$, at a point satisfying (14), with $f_r > 0$ for all $r \in [i, m]$. Put $d_r := f_r$ for $r < l$ and $d_r := 0$ for $r > k$. Then $d_p/f_p \geq d_q/f_q$ for all $p < q$ in $[i, m]$; equivalently $d_p b_q \geq d_q b_p$.

Proof. Only $l \leq p < q \leq k$ needs proof. Let $D = Y_T - g_T$, $\tilde{B}(x) = Y_B + \sum_{r \geq l} \alpha_r b_r/(\alpha_r - x)$ and $F = D - \tilde{B} = U_c - \sum_r \alpha_r f_r/(\alpha_r - x)$. By (14), $D(\theta_{c,r}) = \tilde{B}(\theta_{c,r})$, equal to 0 for interior r and ≥ 0 for $r \in \{l - 1, k\}$. Suppose $d_p/f_p < \lambda < d_q/f_q$ with λ different from every d_s/f_s, so $0 < \lambda < 1$, and put $H = (1 - \lambda)D + \lambda\tilde{B} = D - \lambda F = K_0 - \sum_{s=i}^{m} \alpha_s c_s/(\alpha_s - x)$ with $c_s = d_s - \lambda f_s \neq 0$. The signs σ_s of c_s are $+$ for $s < l$, $-$ for $s > k$, $\sigma_p = -$, $\sigma_q = +$; the sequence $\sigma_i, \ldots, \sigma_m$ has $C \geq 3$ sign changes, all between $s = l - 1$ and $s = k + 1$. Write $H = P(x)/\prod_s(\alpha_s - x)$, $\deg P \leq n := m - i + 1$, $P \not\equiv 0$.

Near α_s, $H \approx -\alpha_s c_s/(\alpha_s - x)$, so on $I_r = (\alpha_{r+1}, \alpha_r)$, $i \le r < m$, H has sign σ_{r+1} at the left end and $-\sigma_r$ at the right end. If $\sigma_r = \sigma_{r+1}$, P has a zero in I_r. If $\sigma_r \ne \sigma_{r+1}$, then $l-1 \le r \le k$, both ends of I_r have the same sign, and $H(\theta_{c,r}) = 0$ (r interior) or $H(\theta_{c,r}) \ge 0$ with both ends negative ($r = l-1$: ends $\sigma_l = -$ and $-\sigma_{l-1} = -$; $r = k$: ends $\sigma_{k+1} = -$ and $-\sigma_k = -$); either way P has at least two zeros in I_r, counted with multiplicity. In total P has at least $(n-1-C)+2C \ge n+2$ zeros, a contradiction. □

Lemma 32 below is the case $(p,q) = (\mathrm{B},\mathrm{C})$ of the feed column of K, with a direct proof that does not need (N).

Light and heavy parts. A function $y(x) = U - \sum_p \alpha_p w_p/(\alpha_p - x)$ with $w \ge 0$ (nonincreasing on I_j) is written $y = U - G_y + K_y$ with $G_y(x) = \sum_{p \le j} \alpha_p w_p/(\alpha_p - x)$ (nondecreasing on I_j) and $K_y(x) = \sum_{p>j} \alpha_p w_p/(x - \alpha_p)$ (nonincreasing). This applies to the feed function $a_c := \Lambda_{c,F}$, to $\Lambda_{c,T}$, to $-\Lambda_{c,B}$ and to the fluxes carried by the links into a mixed column: $y_e = \Lambda_{P,T} - \kappa_e$ for a distillate link from P, $y_e = -\Lambda_{P',B} + \rho_e$ for a residue link from P' (κ_e, ρ_e the duties of a condenser or reboiler on the link), so that $a_c = \sum_e y_e$ as functions on I_j (cf. (21)). Under (N) each of them has a unique zero ζ_y in I_j; let θ^c be the zero of a_c and

$$\Gamma(y) := G_y(\zeta_y) - G_y(\theta_j).$$

For the feed column $\Gamma(a) = 0$, since $\zeta = \theta_j$. A mixed column is of type (a) if its distillate is light and its residue heavy, (b) light and mixed, (c) mixed and heavy, (d) mixed and mixed; in type (d), j is an interior root and $\Lambda_{c,T}(\theta^c) = \Lambda_{c,B}(\theta^c) = e_c := V_T - Y_T \ge 0$.

Lemma 28 (local inequality)**.** Under (N), let c be a mixed column, y_1 and y_2 the fluxes of its links to its mixed distillate and residue (when these are mixed), κ the condenser duty on the first of these links (if any), and $\ell_c = A_c$ if c is of type (a) or (b), $\ell_c = 0$ otherwise. Then $\ell_c + \kappa + \sum_e \Gamma(y_e) \ge \Gamma(a_c)$.

Proof. Let $G = G_{a_c} = G^T + G^B$, with G^T, G^B the light parts of $\Lambda_{c,T}$ and $-\Lambda_{c,B}$, and $s_c = \Lambda_{c,T}(\theta^c) = \Lambda_{c,B}(\theta^c) \ge 0$ (Lemma 19). (a) The top section carries light components only: $\Lambda_{c,T} = V_T - G$, so $\ell_c = s_c + G(\theta^c) - G(\theta_j) = s_c + \Gamma(a_c)$. (b) The top carries only light components, the bottom all heavy ones. With $z = \zeta_{y_2}$ ($\Lambda_{c,B}(z) = \rho$), $\ell_c + \Gamma(y_2) - \Gamma(a_c) = s_c - [G^B(\theta^c) - G^B(z)]$. If $z \ge \theta^c$ the bracket is ≤ 0. If $z < \theta^c$, $s_c - \rho = \Lambda_{c,B}(\theta^c) - \Lambda_{c,B}(z) = [G^B(\theta^c) - G^B(z)] + [K^B(z) - K^B(\theta^c)]$ and the second bracket is ≥ 0. (c) Symmetric: with $z = \zeta_{y_1}$, $\kappa + \Gamma(y_1) - \Gamma(a_c) = \kappa + G(z) - G(\theta^c)$; if $z < \theta^c$, $\kappa - s_c = \Lambda_{c,T}(z) - \Lambda_{c,T}(\theta^c) \ge G(\theta^c) - G(z)$. (d) With $e = e_c$, $z_1 = \zeta_{y_1}$ ($\Lambda_{c,T}(z_1) = \kappa$), $z_2 = \zeta_{y_2}$ ($\Lambda_{c,B}(z_2) = \rho$), the claim reads $X := [G^T(z_1) - G^T(\theta^c)] + \kappa - [G^B(\theta^c) - G^B(z_2)] \ge 0$. If $\kappa \le e$ and $\rho \le e$, then $z_1 \ge \theta^c \ge z_2$ and

$$e - \kappa = (z_1 - \theta^c) \sum_p x_p, \quad x_p = \frac{\alpha_p d_p}{|\alpha_p - \theta^c||\alpha_p - z_1|},$$

$$e - \rho = (\theta^c - z_2) \sum_p y_p, \quad y_p = \frac{\alpha_p b_p}{|\alpha_p - \theta^c||\alpha_p - z_2|},$$

$G^T(z_1) - G^T(\theta^c) = \sigma_1(e - \kappa)$ and $G^B(\theta^c) - G^B(z_2) = \sigma_2(e - \rho)$, where $\sigma_1 = X_L/(X_L + X_H)$, $\sigma_2 = Y_L/(Y_L + Y_H)$, $X_L = \sum_{p \le j} x_p$, $X_H = \sum_{p>j} x_p$, and similarly for Y. For $p \le j < q$, $x_p y_q - x_q y_p$ is a positive multiple of $d_p b_q\, v_p^1 v_q^2 - d_q b_p\, v_q^1 v_p^2$ with $v_s^t = 1/|\alpha_s - z_t|$; since $v_p^1/v_p^2 = (\alpha_p - z_2)/(\alpha_p - z_1) \ge 1 \ge (z_2 - \alpha_q)/(z_1 - \alpha_q) = v_q^1/v_q^2$ and $d_p b_q \ge d_q b_p$ (Lemma 27), it is ≥ 0; summing, $X_L Y_H \ge X_H Y_L$, i.e. $\sigma_1 \ge \sigma_2$. Hence $X = \sigma_1 e + (1 - \sigma_1)\kappa - \sigma_2(e - \rho) \ge (\sigma_1 - \sigma_2)e \ge 0$. If $\kappa > e$: $z_1 < \theta^c$ and

$\kappa - e = \Lambda_{c,T}(z_1) - \Lambda_{c,T}(\theta^c) \geq G^T(\theta^c) - G^T(z_1)$, so the first two terms of X add up to at least e, while the last bracket is $\leq e - \rho \leq e$ if $\rho \leq e$ and ≤ 0 if $\rho > e$. If $\kappa \leq e < \rho$: the first bracket is ≥ 0 and the last ≤ 0. □

Theorem 29 (merge-free mixed networks; Agent G)**.** In the permissive model, under (N), if no mixed column of a regular-column configuration has two producers (no *merge node* in the mixed sub-network for j), then $R \geq H_j$.

Proof. The mixed columns form a tree rooted at the feed column; each non-root mixed column c' has a single link e, and $a_{c'} = y_e$. Let $L(c)$ be the sum of ℓ over the mixed columns of the subtree of c plus the condenser duties on the links inside it. From the leaves up, Lemma 28 and $L(c') \geq \Gamma(a_{c'}) = \Gamma(y_e)$ give $L(c) \geq \Gamma(a_c)$. At the root, $\sum_{\text{light exits}} \Lambda_s(\theta_j) + Q(\text{mixed}) = L([1, N]) \geq 0$; conclude with (18). □

For regular-column configurations satisfying (N), Theorem 29 contains Lemma 25 and Lemma 33. Agent G also shows that $R \geq H_j$ holds at every permissive point at which no merge node descends from a type-(d) column whose excess satisfies $\Lambda_{c,T}(\theta_j) > 0$ and $\Lambda_{c,B}(\theta_j) > 0$ (possible only if $e_c > 0$), by combining the sign rule (with superadditivity of $\min(0, \cdot)$ at merges) above those columns and Lemma 28 below them (`general/general-N.md`, Theorem G6). The mechanism that defeats a general proof: Γ is not superadditive at merge nodes (mixing a vapor-rich and a liquid-rich feed lowers the threshold vapor).

13.3 A five-component counterexample in the permissive model

This subsection was added by the independent verifier. The bound $R \geq V_{\text{FTC}}$ is false in the permissive model for $N = 5$. Take the regular-column configuration (basic configuration 161 of the Shah–Agrawal space, all transfers by thermal coupling, B, C and D withdrawn as liquid side draws)

$$\begin{aligned}&\text{ABCDE} \to \text{ABCD/BCDE}, \quad \text{ABCD} \to \text{ABC/BCD}, \quad \text{BCDE} \to \text{BCD/CDE},\\ &\text{ABC} \to \text{AB/BC}, \quad \text{BCD} \to \text{BC/D}, \quad \text{CDE} \to \text{C/DE},\\ &\text{AB} \to \text{A/B}, \quad \text{BC} \to \text{B/C}, \quad \text{DE} \to \text{D/E},\end{aligned}$$

a saturated-vapor feed ($q = 0$),

$$\alpha = \left(\tfrac{10675204}{742583}, \tfrac{5824255}{854876}, \tfrac{529647}{171259}, \tfrac{1673613}{789461}, 1\right) \approx (14.3758,\ 6.8130,\ 3.0927,\ 2.1199,\ 1),$$

and the rational feed $z \approx (0.19036, 0.18832, 0.17089, 0.03888, 0.41156)$ chosen so that the feed roots are the rationals $\theta \approx (12.3250, 5.5538, 2.5300, 1.9504)$ (exact values in the file `permissive_counterexample_N5.json` of the folder `verification/`). Then $H \approx (0.33442, 0.32912, 0.46994, 0.43302)$ and $V_{\text{FTC}} = H_3 \approx 0.469941$, while the permissive model has a feasible point with $R \approx 0.469685$, i.e. $R/V_{\text{FTC}} \approx 0.99946$, at which every net flow is at least 10^{-6} (so assumption (N) holds) and every inequality of (14) holds with slack at least 10^{-6}. That an exact feasible point exists within 10^{-30} of the computed one is proved by a Krawczyk test in interval arithmetic on a square subsystem of the equalities (`verification/krawczyk_certificate.py`); the inequalities and $R < V_{\text{FTC}}$ are then checked over the whole box in interval arithmetic.

The mechanism is the one isolated in Section 13.2: for $j = 3$ the column BCDE is of type (d) and runs about 0.006 above its threshold ($V_T \approx 0.03015$, $Y_T \approx 0.02420$) although C and D distribute in it with positive flows, so its root at $j = 3$ is active and

an infinite-stage column could not carry this excess; its distillate feeds the merge node BCD, which is not the bottom of a simple diamond. In the strict model, Theorem 22 excludes such a point, and a multistart minimization of the same configuration at the same data gives $R/V_{\mathrm{FTC}} = 1$ to 10^{-7}. Hence the permissive (TG-literal) model is not a faithful model for the conjecture, and of the 20 open (basic configuration, j) pairs at least one, (161, 3), is settled negatively.

Physical meaning. We do not know an adiabatic arrangement with pure products that realizes this point, and the natural candidates fail. (i) An infinite-stage column in which C and D appear in both products has an active root and runs at its Underwood vapor (Halvorsen's rule quoted in Section 9.2); excess vapor at a fixed product distribution requires fewer stages, but a section with finitely many stages cannot remove B from the residue or E from the distillate completely, and as the impurities tend to zero both sections become infinite and the excess is forced to zero. So the point is at best the limit of finite-stage operations with *impure* intermediate streams, which the conjecture does not compare. (ii) A vapor bypass carrying the excess from the node below BCDE to the node above it, or a separate section with $V = L$ between these nodes, transports components (E upward into the node BCD, B downward, or C and D in opposite directions), whereas the permissive excess leaves every net flow unchanged; it therefore realizes a different point, outside Definition 13. The excess-vapor device of TG is physically sound for columns in which at most one component distributes (then V above the threshold does not change the distribution), and the strict model keeps exactly that freedom. We therefore regard the point as an artifact of the permissive formulation, not as a counterexample to a physical reading of the conjecture; it does, however, show that a global optimizer applied to TG's model can return configurations below the FTC value. Version 2 proves that no family of steady states of any arrangement of $\mathcal{A}$ with purities tending to 1 approaches this point (Section 4).

13.4 The FTC configuration, every *N*

Theorem 30 (minimum vapor of the generalized Petlyuk arrangement)**.** In both models, every feasible point of the FTC configuration satisfies $R \geq \max_j H_j$.

Proof. Fix j. The mixed streams form the grid (i, m), $1 \leq i \leq j < m \leq N$; the column (i, m) splits $[i, m]$ into $[i, m-1]$ and $[i+1, m]$. Row i is the chain $(i, N) \to (i, N-1) \to \cdots \to (i, j+1)$ of distillate links, ending in the light exit of $(i, j+1)$; the column (i, m) also receives the flux $x_m^{(i)} := -B_{(i-1,m)}$ from row $i-1$ ($x^{(1)} \equiv 0$; all links are couplings or side draws, so (21) has no duty terms). Put $y_m^{(i)} = -B_{(i,m)}$ and $Y_m^{(i)} = \sum_{m' \geq m} y_{m'}^{(i)}$, $X_m^{(i)} = \sum_{m' \geq m} x_{m'}^{(i)} = Y_m^{(i-1)}$.

Claim: if $X_m^{(i)} \leq 0$ for all m, then $Y_m^{(i)} \leq 0$ for all m. Along the row, $a_{(i,N)} = x_N$ and $a_{(i,m)} = x_m + A_{(i,m+1)}$. We show $A_{(i,m)} \geq X_m$ by descending induction: $A_{(i,N)} \geq \min(0, x_N) = X_N$; if $A_{(i,m+1)} \geq X_{m+1}$ then $a_{(i,m)} \geq X_m$ and $A_{(i,m)} \geq \min(0, a_{(i,m)}) \geq \min(0, X_m) = X_m$ (Lemma 19). Since $y_m = a_{(i,m)} - A_{(i,m)}$, telescoping gives $Y_m = X_m - A_{(i,m)} \leq 0$.

Row 1 has $X \equiv 0$ (the feed column has $a = 0$). By induction $Y_{j+1}^{(j)} \leq 0$; but $-Y_{j+1}^{(j)}$ is the sum of Λ over the heavy exits (the bottom sections of the columns (j, m)). The second form of (18) gives $R - H_j \geq -Y_{j+1}^{(j)} \geq 0$. □

This proves, in the Underwood model, the statement of [1] §3.1 and §5 ("the minimum-energy solution for a generalized Petlyuk arrangement ... is given by the highest peak in the $V_{\min}$ diagram") for every N, the attainment being Theorem 35. When a grid node is missing, the rows no longer absorb the residue fluxes of the previous row, and the induction breaks; family K is the smallest instance.

13.5 Four components

For $N = 4$, $j \in \{1, 3\}$ is Theorem 26. For $j = 2$ the mixed streams are among ABCD, ABC, BCD, BC (A, B, C, D = components 1–4). ABC is present iff it is the distillate of ABCD, BCD iff it is its residue, and BC, if present, is the residue of ABC and the distillate of BCD whenever these are present. Hence: if ABC or BCD is absent, the mixed streams form a path (Lemma 25); if ABC, BCD and BC are present, Lemma 31; the remaining family is

$$\begin{aligned}\mathrm{K} = \{&\mathrm{ABCD} \to \mathrm{ABC/BCD},\ \mathrm{ABC} \to \mathrm{AB/C},\ \mathrm{BCD} \to \mathrm{B/CD},\\ &\mathrm{AB} \to \mathrm{A/B},\ \mathrm{CD} \to \mathrm{C/D}\}\end{aligned}$$

with its 16 coupling/exchanger variants (B and C are liquid side draws).

Lemma 31 (diamond)**.** If ABC, BCD and BC are present ($N = 4$, $j = 2$), then $R \geq H_2$ in both models.

Proof. Columns 0 = ABCD, 1 = ABC, 2 = BCD, 3 = BC; light exits: tops of 1 and 3. $a_0 = 0$ gives $A_0 = B_0 \geq 0$; $a_1 = A_0 - \kappa_1 \geq -\kappa_1$, $a_2 = -B_0 + \rho_2$, $a_3 = -B_1 + A_2$ (side draw). By Lemma 19, $A_1 \geq \min(0, a_1) \geq -\kappa_1$. If $a_3 \geq 0$ then $A_3 \geq 0$ and $A_1 + A_3 \geq -\kappa_1$. Otherwise $A_3 \geq a_3$ and $A_1 + A_3 \geq A_1 - B_1 + A_2 = a_1 + A_2$; if $a_2 \geq 0$ then $A_2 \geq 0$ and the sum is $\geq -\kappa_1$; if $a_2 < 0$ then $A_2 \geq a_2$ and the sum is $\geq A_0 - \kappa_1 - B_0 + \rho_2 \geq -\kappa_1$. Conclude with (18), $Q(\text{mixed}) = \kappa_1$. □

Lemma 32 (recovery ordering)**.** If the feed column splits ABCD into ABC/BCD (permissive model), its top recoveries satisfy $d_B/z_B > d_C/z_C$. In particular $d_B > 0$ and $b_C = z_C - d_C > 0$.

Proof. The roots of the feed column are $\theta_1, \theta_2, \theta_3$, θ_2 interior. By (14), $\Phi_1 := g_T(\theta_2) - g_T(\theta_1) \geq 0$ and $\Phi_3 := g_T(\theta_2) - g_T(\theta_3) \geq 0$, with $d = (z_A, d_B, d_C, 0)$. On the diagonal $d_B = r z_B$, $d_C = r z_C$ ($0 \leq r \leq 1$), the feed equation gives $g_T(\theta_t) = r(1 - q) + (1 - r)\mathcal{A}(\theta_t) + r\mathcal{D}(\theta_t)$ with $\mathcal{A}(x) = \alpha_A z_A/(\alpha_A - x)$ increasing and $\mathcal{D}(x) = \alpha_D z_D/(x - \alpha_D)$ decreasing on (α_D, α_A). Thus $\Phi_1 = -(1-r)A_{12} + rD_{12}$ and $\Phi_3 = (1-r)A_{32} - rD_{32}$ with $A_{12} = \mathcal{A}(\theta_1) - \mathcal{A}(\theta_2)$, $D_{12} = \mathcal{D}(\theta_2) - \mathcal{D}(\theta_1)$, $A_{32} = \mathcal{A}(\theta_2) - \mathcal{A}(\theta_3)$, $D_{32} = \mathcal{D}(\theta_3) - \mathcal{D}(\theta_2)$, all positive, and

$$\frac{A_{12}D_{32}}{A_{32}D_{12}} = \frac{(\alpha_A - \theta_3)(\theta_1 - \alpha_D)}{(\alpha_A - \theta_1)(\theta_3 - \alpha_D)} > 1.$$

For $r = 0$, $\Phi_1 < 0$; for $r = 1$, $\Phi_3 < 0$; for $0 < r < 1$, $\Phi_1, \Phi_3 \geq 0$ would give $A_{12}D_{32} \leq A_{32}D_{12}$. So no feasible point lies on the diagonal. Moreover $\partial\Phi_1/\partial d_B = \alpha_B/(\alpha_B - \theta_2) + \alpha_B/(\theta_1 - \alpha_B) > 0$ and $\partial\Phi_3/\partial d_B = \alpha_B/(\alpha_B - \theta_2) - \alpha_B/(\alpha_B - \theta_3) > 0$: lowering d_B below $z_B d_C/z_C$ lowers both, so $d_B/z_B \leq d_C/z_C$ is infeasible. As the inequality is strict, $d_B > 0$ and $d_C < z_C$. □

Lemma 33 (family K)**.** For every variant of family K, $R \geq H_2$ in both models.

Proof. Columns 0 = ABCD, 1 = ABC, 2 = BCD; light exits: tops of 1 and 2. Since θ_2 is an interior root of column 0, $A_0 = B_0 = V_T - Y_T =: e \geq 0$. Column 1 is fed with $d = (z_A, d_B, d_C)$ and $\Lambda_{1,F} = \Lambda_{0,T} - \kappa_1$; its split AB/C has the single root $\rho_1 := \theta_{1,2} \in I_2$, with $\Lambda_{1,T}(\rho_1) \geq 0$. Column 2 is fed with $b = (b_B, b_C, z_D)$, $\Lambda_{2,F} = -\Lambda_{0,B} + \rho_2'$ (ρ_2' the reboiler duty on the link, 0 for a coupling), root $\rho_2 := \theta_{2,2}$, $\Lambda_{2,T}(\rho_2) \geq 0$. Let $\Delta_p(x) = 1/(\alpha_p - x) - 1/(\alpha_p - \theta_2)$, whose sign is that of $x - \theta_2$ for every p. Then

$$A_1 \geq \sum_{p=A,B} \alpha_p d_p \Delta_p(\rho_1), \quad e - \kappa_1 = \sum_{p=A,B,C} \alpha_p d_p \Delta_p(\rho_1),$$
$$A_2 \geq \alpha_B b_B \Delta_B(\rho_2), \qquad \rho_2' - e = \sum_{p=B,C,D} \alpha_p b_p \Delta_p(\rho_2),$$

the equalities because the feed functions vanish at ρ_1, ρ_2 and equal $e - \kappa_1$, $-e + \rho_2'$ at θ_2. By (18) it suffices to show $A_1 + A_2 + \kappa_1 \geq 0$. If $\rho_1, \rho_2 \geq \theta_2$: $A_1, A_2 \geq 0$. If $\rho_1 \leq \theta_2 \leq \rho_2$: $A_2 \geq 0$ and $A_1 + \kappa_1 \geq e - \alpha_C d_C \Delta_C(\rho_1) \geq e$. If $\rho_1, \rho_2 \leq \theta_2$: $A_1 + \kappa_1 \geq e$ and $A_2 \geq \rho_2' - e - \alpha_C b_C \Delta_C(\rho_2) - \alpha_D z_D \Delta_D(\rho_2) \geq -e$. In the remaining case $\rho_1 \geq \theta_2 \geq \rho_2$ put

$$X = \alpha_C d_C \Delta_C(\rho_1),\ Y = \alpha_B d_B \Delta_B(\rho_1),\ Z = \alpha_B b_B |\Delta_B(\rho_2)|,\ W = \alpha_C b_C |\Delta_C(\rho_2)|,$$

all ≥ 0, with $X + Y \leq e - \kappa_1 \leq e$, $Z + W \leq e - \rho_2' \leq e$ and $A_1 + A_2 + \kappa_1 \geq e - X - Z$. Now $X/(X+Y) + Z/(Z+W) - 1 = (XZ - YW)/((X+Y)(Z+W))$, so $XZ \leq YW$ implies $X + Z \leq e$ (if $X + Y = 0$ or $Z + W = 0$ this is immediate). Finally $XZ \leq YW$ reads

$$\frac{d_C b_B}{d_B b_C} \leq \frac{\mathrm{Rat}(\rho_2)}{\mathrm{Rat}(\rho_1)}, \qquad \mathrm{Rat}(x) := \frac{\Delta_C(x)}{\Delta_B(x)} = \frac{(\alpha_B - x)(\alpha_B - \theta_2)}{(x - \alpha_C)(\theta_2 - \alpha_C)}.$$

The left side is ≤ 1 by Lemma 32 ($d_B z_C \geq d_C z_B \iff d_B b_C \geq d_C b_B$); Rat is decreasing on I_2 and $\rho_1 \geq \rho_2$, so the right side is ≥ 1. (By Lemma 32, $d_B > 0$ and $b_C > 0$, so $\rho_1 < \alpha_B$ and $\rho_2 > \alpha_C$ and all quantities are finite.) □

Theorem 34 ($N = 4$)**.** In both models, every feasible point of every one of the 152 four-component configurations satisfies $R \geq \max(H_1, H_2, H_3)$.

Proof. Theorem 26, Lemmas 25, 31 and 33, by the case distinction above. □

Agent G also proves, for every N and under (N), that $R \geq H_j$ in the permissive model whenever every merge node of the mixed sub-network is the bottom of a *simple diamond*: a type-(d) column whose distillate column is of type (b) and whose residue column is of type (c), both feeding the merge node (`general/general-N.md`, Theorem G8). With Theorem 29 this gives an independent proof of Theorem 34 under (N); the proof above needs no non-degeneracy assumption.

14 Attainment by the FTC

Theorem 35. For every N, α, $z > 0$ and $q \in \mathbb{R}$, the FTC configuration has an operating point, feasible in the strict (hence in the permissive) model, with $R = \max_j H_j$; all its flows are positive.

Proof. *Preferred split.* Let a column $[i,m]$, $m-i\geq 2$, have positive feed f and net vapor U such that $\theta_i,\dots,\theta_{m-1}$ are roots of its feed equation. Define V_T and d by

$$\alpha_p d_p = V_T\,\frac{\prod_{r=i}^{m-1}(\alpha_p-\theta_r)}{\prod_{p'\in[i,m-1],p'\neq p}(\alpha_p-\alpha_{p'})}\qquad (i\leq p\leq m-1),$$

where the factor V_T is fixed by the condition $d_i=f_i$ (the formula for $p=i$), and $b=f-d$ ($d_m:=0$), $V_B=V_T-U$. Partial fractions give $V_T-g_T(x)=V_T\prod_r(x-\theta_r)/\prod_{p=i}^{m-1}(x-\alpha_p)$, so $g_T(\theta_r)=V_T$ for all r; with the feed equation, $g_B(\theta_r)=V_B$, and since V_B-g_B has its poles at $\alpha_{i+1},\dots,\alpha_m$, $V_B-g_B(x)=V_B\prod_r(x-\theta_r)/\prod_{p=i+1}^{m}(x-\alpha_p)$ ($V_B\neq 0$, otherwise $b\equiv 0$, contradicting $b_m=f_m>0$). Counting signs of the factors, every d_p has the sign of V_T and every b_p the sign of V_B; as $d_i=f_i>0$ and $b_m=f_m>0$, all are positive. At $x=0$: $L_T=V_T-\sum d=V_T\prod_r\theta_r/\prod_{p<m}\alpha_p$ and $L_B=V_B+\sum b=V_B\prod_r\theta_r/\prod_{p>i}\alpha_p$, so $0<L_T<V_T$ and $L_B>V_B>0$. The column satisfies (14) with $Y_T=V_T$ and all roots active.

Construction. Process the columns by decreasing length. The feed column ($f=z$, $U=1-q$) and every column $[i,m]$ with $m-i\geq 2$ are put at the preferred split. A child $[i,m]$ receives d from $[i,m+1]$ and b from $[i-1,m]$ (those present), with $U=V_T[i,m+1]-V_B[i-1,m]$; since both producers have every common root active, $\theta_i,\dots,\theta_{m-1}$ are roots of its feed equation (carry-over [11] [12]), and its feed is positive. A binary column $[r,r+1]$ gets $d_r=f_r$, $b_{r+1}=f_{r+1}$, $V_T=\alpha_r f_r/(\alpha_r-\theta_r)+e_r$, $V_B=\alpha_{r+1}f_{r+1}/(\theta_r-\alpha_{r+1})+e_r$ with $e_r:=\max_s H_s-H_r\geq 0$; it has no interior root, so excess vapor is allowed in both models.

Side draws. It remains to check $V_T[p,p+1]=V_B[p-1,p]$ at the side draws $p=2,\dots,N-1$ (product purities hold by construction). Let δ_p be the vapor lost at the node of p (inflow from below minus outflow above), treated in Proposition 16 as a condenser duty (or a reboiler duty $-\delta_p$). For the split r, the light exits are the tops of $[i,r+1]$, $i\leq r$; for $i<r$ the column is at its preferred split with root θ_r active, so $\Lambda=0$; for $i=r$, $\Lambda=e_r$. The only other heat duty is the reboiler of product N. Proposition 16 with $U=$ light streams gives $R-\sum_{p=r+1}^{N-1}\delta_p=H_r+e_r=\max_s H_s$ for $r=1,\dots,N-1$. For $r=N-1$ this is $R=\max_s H_s$, and successive differences give $\delta_{N-1}=\cdots=\delta_2=0$. □

The construction is the one of [1] §3 (every internal column at its preferred split); the excess vapor is placed in the binary columns, whose children are pure products, so that no carried-over root moves.

15 Version 1: computer verification and numerical evidence

No proof above depends on a computation. The scripts check the algebra, cross-check the case analysis and search for counterexamples (environment: WSL Fedora, Python 3 with sympy, mpmath, scipy, pyomo, ipopt; paths relative to the problem folder).

- `theory/scripts/check_algebra.py` (sympy, exact): the identities used in Lemmas 32, 33 and Theorem 35. PASS.

- `theory/scripts/check_ftc_achievability.py` (mpmath, 80 digits): builds the point of Theorem 35 for 200 random instances for each $N=3,\dots,8$ ($q\in[-1,2]$) and checks every constraint of the strict model, the side draws and $R=\max H_j$ to 10^{-50}. PASS.

- `theory/scripts/signcheck.py certify4`: decides, by linear programming over the sign patterns of the inflows, which (configuration, j) pairs are not settled by Lemma 19, (21) and the cut identity alone; for $N = 4$ these are exactly the 16 variants of family K at $j = 2$. PASS. (For $N = 5$ the sign rule alone leaves 1312 (configuration, j) pairs for each of $j = 2, 3$, in 2160 of the 6128 configurations and 48 basic configurations. The LPs have integer data but are solved in floating point (HiGHS); the statement that the other pairs are settled by the sign rule is therefore computer-assisted, not a hand proof.)

- Agent C (`computation/`): the enumeration of the configuration space (8/152/6128 configurations, as in [7]); an exact-arithmetic check of the cut identity on all configurations for $N \le 5$; an exact enumeration of sign-consistent light-to-heavy paths (none in the strict model for $N \le 6$; family K only for $N = 4$ in the permissive model); multistart local minimization (ipopt, every accepted point re-verified a posteriori) of R in the strict model over all 152 configurations at 239 random points for $N = 4$, $q = 1$ (minimum ratio $R/V_{\mathrm{FTC}} = 1 - 1.4 \cdot 10^{-9}$), and over all 6128 configurations at the eight test points of [4] for $N = 5$ (minimum $= V_{\mathrm{FTC}}$ to 10^{-10} relative, the FTC attaining it); adversarial joint minimization over (α, z, q) and the operating point (ratio never below 1 beyond 10^{-6}), and targeted minimization on the permissive "bad paths" (family K; the all-coupled variants of the 48 five-component families; 181 of the merge cases), never below 1; SCIP global optimization used only as a cross-check at two points for $N = 4$ (strict model, all 152 configurations, at $\alpha = (8, 4, 2, 1)$, $z_p = 1/4$, $q = 1$: SCIP and ipopt agree to $2.2 \cdot 10^{-9}$, and the SCIP dual bound is $\ge V_{\mathrm{FTC}}$ for every non-tying configuration, while for the tying ones it reached $V_{\mathrm{FTC}}(1 - 8 \cdot 10^{-6})$ at the time limit; permissive model, first 100 configurations, at the point of the example below: agreement to $3.7 \cdot 10^{-9}$). Many exact ties (for $N = 4$, $q = 1$, 12 to 16 configurations tie with the FTC, depending on the dominant peak; 2717 of 2720 observed ties satisfy Corollary 23). Random searches with $q \ne 1$ and a random permissive search for $N = 5$ were not run. The FTC value reproduces the eight five-component optima of [4] (all eight test points have an extreme highest peak, $j = 1$ or $j = 4$, so they are covered by Theorem 26 in both models), and the model reproduces Halvorsen's ternary table [6] (Table 6.1). Scripts and data: `computation/scripts/` and `computation/certificates/`.

- Agent G (`general/scripts/`): `recovery_order_lp.py` (LP minimum of $d_p/f_p - d_{p+1}/f_{p+1}$ over the constraints (14) of random columns, 1587 LPs, positive), `classify_mixed.py` (types and merge nodes of the mixed sub-networks for $N = 4, 5, 6$), `gamma_check.py` (stress test of Lemmas 19, 27, 28 at optimal and random feasible points). `theory/scripts/opencount.py` intersects the sign-hard pairs with the pairs having a merge node ($N = 5$: 864 pairs, 720 configurations; Agent C's path enumeration gives the same 20 (basic configuration, j) pairs).

- `theory/scripts/stressK.py`: minimizes $d_B/z_B - d_C/z_C$, $A_1 + A_2 + \kappa_1$ and $R - H_2$ over the 16 variants of K in the permissive model: minima $+0.039$, $-2 \cdot 10^{-10}$ and $-2 \cdot 10^{-10}$ (round-off), consistent with Lemmas 32 and 33 and showing that the bound is tight.

- Independent verification (`verification/`, code written from the equations, sharing nothing with the scripts above): an exact (rational) check of the node-balance form of the cut identity on 400 random networks of Definition 13 (several columns per label,

merges, heaters and coolers at every node, random subsets U, rational evaluation points); an independent construction of the attainment point of Theorem 35 (60 digits, $N = 3, \dots, 7$, 200 instances); the FTC value at the eight test points of [4]; an independent enumeration (3/8, 18/152, 203/6128) and minimum-vapor NLP for all 8 ternary and all 152 four-component configurations (13 runs in the strict or the permissive model) at 9 parameter points ($q \in \{-0.3, 0.3, 0.4, 0.5, 1, 1.3\}$, four of them with the middle peak highest): Halvorsen's Table 6.1 is reproduced (1.882, 2.032, 2.072, 1.556, 1.366), no configuration falls below V_{FTC} beyond the feasibility tolerance (10^{-7}), and all 120 observed ties satisfy Corollary 23. For $N = 5$, a multistart probe of the 20 open (basic configuration, j) pairs in the permissive model (all-coupled and all-exchanger variants, 220 random parameter points with the peak H_j highest and the others within 10%, $q \in \{-0.3, 0, 0.3, 0.5, 1, 1.2\}$) found one point below V_{FTC}, for the pair (161, 3); after re-optimization with margins it is certified by `krawczyk_certificate.py` (Section 13.3). All other probe values are $\geq 1 - 1.5 \cdot 10^{-7}$.

Example 36 (the identity at work). $N = 4$, $\alpha = (47.5774, 25.5263, 15.1071, 1)$, $z = (0.1358, 0.5169, 0.0923, 0.255)$, $q = 1$ (Agent C, `computation/scripts/cut_identity.py`, local optima of the strict model). The peaks are $H = (0.93322, 1.588798, 0.786044)$, so $V_{\mathrm{FTC}} = H_2 = 1.588798$. The FTC and the all-coupled variant of family K both reach $R = 1.588798$, every term of (18) vanishing at $j = 2$. The K variant with heat exchangers on ABC, BCD and AB has $R = 1.647249$ and $R - H_2 = 0.058451 = 0.021470\,(Q) - 0.002772\,(\Lambda$, top of ABC$) + 0.039753\,(\Lambda$, top of BCD): a negative light-exit flux is compensated. The indirect sequence has $R = 2.481686$. ⌟

16 Version 1: scope as stated in version 1

The items below are those of version 1. For the lower bound, items 1–3 and the questions of reverse flows and degenerate points are settled by version 2 (Section 4).

(1) *Class.* Theorem 22 and Theorem 26 hold for every two-section column network (Definition 13); Theorem 34 is proved for the 152 regular-column configurations with $N = 4$. The wording "any adiabatic distillation arrangement" of [1] would also admit columns with several feed stages (intermediate sections between feeds), stream splitters dividing one section product among several columns, recycles, non-interval intermediate streams, splits in which one product keeps every component of the feed, streams of different labels merged at one feed stage, and reverse net flows; none of these is covered. Heat integration between streams and several pressure levels are excluded by the conjecture itself.

(2) *Reverse flows.* Hypothesis (H1) is part of the model of [4] [5]. The cut identity holds without it, but Lemmas 19 and 20 use the monotonicity of Lemma 15, and the conditions (14) themselves rest on Underwood's interlacing of roots, which fails with reverse flows [6]. Whether reverse flows could lower R is open.

(3) *Model.* The theorems are statements about Underwood's infinite-stage model as formulated by TG, with the split of each column fixed by the network (components $l, \dots, k$ nominally distributing, possibly with zero flow on one side); its physical justification is that of [6] [4]. At degenerate points (a nominally distributing component absent from one product) the strict model forbids excess vapor that an infinite-stage column could physically carry, and imposes TG's equalities at roots

that are not active (Section 9.2); physical operating points of this kind are not covered by Theorem 22.

(4) *Permissive model, $N \geq 5$.* The bound can fail only for splits j whose mixed subnetwork has a merge node below a type-(d) column carrying excess vapor, when the sign rule does not settle the case. For $N = 5$ these are 20 (basic configuration, j) pairs, i.e. 864 (configuration, j) pairs and 720 of the 6128 configurations (in 16 basic configurations), and it does fail for the pair (161, 3) (Section 13.3); the other 19 pairs are undecided (a verifier's multistart search on their all-coupled and all-exchanger variants found no other point below V_{FTC}). All other configurations are covered (Theorems 26 and 29, the latter under (N), and the sign rule, the last by the floating-point LP classification of `signcheck.py`). Numerical searches find no counterexample. Degenerate points (assumption (N) violated) are not covered by Theorem 29.

(5) *Products and feed.* Pure products withdrawn as saturated liquid, a single feed (H2). Nonsharp product specifications and several feeds are not treated. (A pure product withdrawn as saturated vapor instead would only increase the bound: in the node balance a heavy vapor product contributes $\theta_j z_p/(\theta_j - \alpha_p)$ instead of $\alpha_p z_p/(\theta_j - \alpha_p)$, and light products do not enter (17).)

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